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Runtime Validity Estimation for Integrating Hybrid Learning-Based Models into Vehicle Dynamics Control

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Abstract

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Chapter 1

Introduction

Consider a driver traveling along a highway at 100 km/h when it starts to rain. As visibility decreases, the driver decides to pull over onto the shoulder. During the maneuver, water on the road surface may reduce tire–road grip, making it more difficult to change direction. Under these conditions, what began as an attempt to stop may lead to a loss of control and a risk of collision with other vehicles or roadside barriers.

The difficulty in this example lies in executing a maneuver when the available grip changes. Steering and braking depend on forces transmitted through tire–road contact, so a change in these conditions may alter the vehicle's ability to follow the intended path [1].

Available grip varies with the pavement surface and the conditions encountered along the route. Pavement friction studies relate skid resistance to surface texture and road characteristics [19, 10]. Water on the road surface may impair tire–road contact, while ruts in the wheel paths may retain water and increase the risk of hydroplaning [15].

These limits become relevant when the driver needs to change lanes with little time to react or swerve to avoid an obstacle. The lateral forces required to follow the intended path may exceed those that tire–road contact can transmit [1]. At high speeds and with low grip, this limitation may cause lateral sliding and yaw instability, compromising the maneuver [9].

Vehicle control systems assist the driver with braking and maintaining stability. The anti-lock braking system (ABS) aims to prevent wheel lock during braking [4]. Electronic stability control (ESC) helps correct deviations between vehicle motion and the direction intended by the driver; Bosch's electronic stability program (ESP), introduced in 1995, is an implementation of this function [3]. Since then, development has extended to coordinating braking, steering, and propulsion. This coordination provides more ways to influence vehicle motion and requires information about its response, including states and parameters that may need to be estimated during operation [16].

1.1 Context and Motivation

The vehicle's response also depends on its mass, load, and tire properties. Friction and axle loads affect braking distance prediction [23], while mass and tire cornering stiffness enter the description of lateral and yaw motion [22]. In the linear range, cornering stiffness is the slope of the relationship between tire lateral force and slip angle [16]. It therefore describes an aspect of tire behavior distinct from the friction limit: the latter bounds the available forces, whereas stiffness influences how lateral force develops with slip angle. Variations in these quantities may affect both the forces available for a maneuver and the resulting motion.

Studies on adaptive vehicle control illustrate the consequences of these variations. Xu et al. evaluate path tracking through simulations on roads with low and variable friction, obtaining smaller lateral and heading errors with their adaptive strategy than with the comparison controllers [20]. Yeneneh et al. use runtime parameter estimation to update a lateral tracking model that includes mass and cornering stiffness. In their simulations, the adaptive controller reduces tracking errors relative to conventional predictive control and a linear quadratic regulator [22]. These results motivate considering changes in both the road environment and the vehicle itself, and examining whether the dynamics description used for control remains representative.

Predictive controllers use a model to relate control inputs to the expected vehicle response. Model predictive control (MPC), including its nonlinear formulation (NMPC), uses these predictions to select inputs over a finite horizon while accounting for constraints [17]. The model must provide sufficient detail to support these decisions while keeping the control problem solvable with the available computational resources. Chen et al., for example, use a simplified bicycle model to balance prediction accuracy and computational cost [7]. These simplifications may introduce discrepancies between predicted and actual motion. Pan et al. identify model structure and parametric errors as sources of these discrepancies and account for them in a robust predictive controller [14].

Uncertainty about a parameter's value does not imply that it is changing. A difference between its actual and assumed values may affect predictions even when its physical value remains constant. Yeneneh and Sufe consider uncertainty in mass, front and rear tire cornering stiffnesses, and road friction within a robust predictive control formulation [21]. Li et al. address uncertainty in cornering stiffness and yaw moment of inertia through a controller with longitudinal and lateral coupling, evaluated in hardware-in-the-loop (HIL) tests [13]. These studies motivate accounting for parametric uncertainty when using a simplified model for control.

It is therefore useful to distinguish the vehicle's physical parameters, the nominal values used by the model, and estimates obtained from measurements during operation. Estimation may reduce parametric error, but it does not eliminate structural model ap-

proximations. Chen et al. reflect this distinction by combining calibration of a nominal physical model with compensation for the remaining unmodeled dynamics [6]. Their formulation illustrates why improving prediction may require addressing discrepancies that parameter calibration alone does not capture.

Hybrid modeling addresses some of these discrepancies by combining physical equations with information learned from data [8, 24]. In a model with an additive learned residual, the physical component provides a nominal prediction and the learned component estimates a correction from the difference between that prediction and the observed response [11, 7]. The correction can represent effects that the simplified model does not describe adequately. In the formulation of Kabzan et al., training targets are obtained from measured state transitions and nominal predictions, allowing model error to be learned without separately identifying its physical causes [11].

Kabzan et al. use Gaussian process regression for this correction in autonomous racing. During tests with a full-scale race car, the corrected model maintained an approximately constant mean prediction error as driving became more demanding over successive laps, whereas the nominal model's error increased [11]. Learning can also be assigned to individual force components. Zhou et al. combine a neural-network-based traction model with physical longitudinal dynamics and an estimator of vehicle mass and a modified rolling friction coefficient. Their evaluation includes loads and road conditions different from those used for training [24]. This is a hybrid force-modeling and parameter-estimation architecture, in which learning plays a different role from that of an additive residual.

Other hybrid approaches fuse physical and neural models to predict dynamics or couple a neural estimator with a physical observer to estimate vehicle sideslip angle [8, 2]. Physical knowledge can also guide training: Wang et al. include force-balance equation residuals in the learning objective of a vehicle dynamics model [18]. These uses of learning serve different purposes. This thesis focuses on an additive residual within the controller's prediction model, evaluated both for prediction accuracy and for its effect on control.

Consider, as a conceptual example, a residual trained with a particular vehicle load and subsequently used on a trip with additional passengers or cargo. If the nominal model and residual remain unchanged, the learned correction may cease to describe the discrepancy under the new load. Changes in tire properties may pose a similar difficulty: familiar speeds and steering inputs need not produce the same response when the underlying parameters differ [21].

Load may vary between trips, and tire characteristics may evolve during driving. Kalaria et al. learn and update a tire model during operation to account for changes in lateral slip behavior; their evaluation uses numerical and high-fidelity simulators and includes simulated changes intended to represent wear [12]. More generally, Brunke et al. identify data scarcity and distribution shifts as practical challenges in learning-based control [5]. A change does not necessarily invalidate a learned residual, but performance

under some conditions does not by itself establish its usefulness under others. Chen et al. identify the limited ability of a correction trained before operation to adapt to new conditions as a motivation for updating it during operation [7].

The research challenge is to assess during operation whether the residual continues to provide useful information for control when the vehicle response changes, and to determine how this assessment should influence the controller. This requires relating model error to the decisions it affects, without necessarily identifying the physical cause of the discrepancy.

1.2 Research Problem

The problem is studied using a control architecture that distinguishes the reference plant from the internal models considered by the controller. CarSim represents the vehicle's high-fidelity response, whereas the MPC must operate with descriptions sufficiently compact to make predictions and optimize steering actions at runtime. The starting point is a nominal linear bicycle model, denoted B0.

The research is not limited to training a single residual model. Several hybrid and supervisory configurations, denoted B1, B2, B3, and B4, are built from B0 and compared to analyze different ways of learning, evaluating, and using the discrepancy between the physical model and the plant. As a basic example, B1 retains the physical structure of B0 and incorporates a neural residual correction learned from CarSim data. Figure 1.1 illustrates this configuration and summarizes the relationship between the reference plant, the hybrid model, and the controller. The description and formulation of B0–B4 will be discussed in Chapter 3, while their parameters, data, and training procedures will be specified in Chapter 7.

Within the domain represented by the training data, residual models may improve the accuracy of B0. However, this improvement does not guarantee that each correction remains useful when the operating distribution changes. Variations in grip, mass or load, and tire cornering stiffness, as well as speeds not represented during training or sensor degradation, may modify the learned relationships between the state, the steering input, and the model–plant discrepancy.

The central problem arises because the MPC may continue using learned information even after its predictive validity has decreased. B1 illustrates the basic case: a residual correction alone does not indicate whether it remains reliable for control decisions. The same problem must be examined in the other configurations, B2–B4. A degraded prediction may alter the optimized action sequence and impair closed-loop performance. Accordingly, the research addresses how to characterize the validity of learned models during operation, how to relate it to control-relevant error, and how to make the MPC respond when this validity decreases.

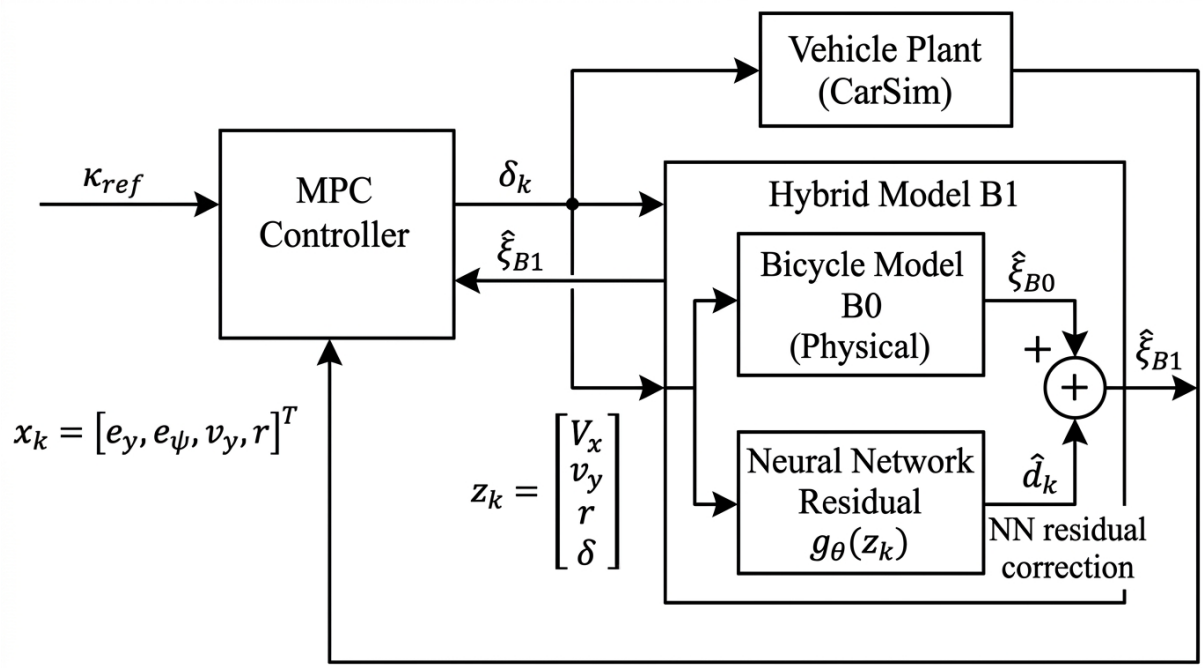


Figure 1.1: Control structure for configuration B1, used as an example of a hybrid model comprising the B0 bicycle model and a neural residual.

1.3 Scope of the Problem

This thesis focuses on vehicle lateral dynamics and path tracking using an MPC/NMPC controller. The prediction model combines a control-oriented physical model with a residual learned from data. Within this architecture, the study will investigate runtime estimation of the validity of the learned correction and its use to adapt the controller when conditions differ from those represented during training. The families of change considered comprise variations in grip, mass or load, and tire cornering stiffness, as well as operation outside the range of states, speeds, or inputs used to train the residual. The methodology will select representative scenarios from these families, without attempting to cover all combinations or necessarily identify the physical parameter responsible for degradation.

The experimental protocol will distinguish measured variables, estimated states or parameters, and quantities unavailable to the controller. The kinematic signals and steering input required by the model will be considered available through measurement or estimation, as appropriate. In contrast, the true values of parameters such as grip, mass, inertias, or cornering stiffness may be used to configure and analyze scenarios without being supplied to the validity estimator or controller when the experiment defines them as unknown. The scope does not include perception, object detection, global planning, prediction of other agents, or guarantees for the complete autonomous system; in particular, the loss of visibility in the opening example is not a perception problem addressed by the thesis. The main contribution will be evaluated through predictive and closed-loop performance and empirical robustness criteria, without promising formal safety or

stability guarantees.

1.4 Research Gap

1.5 Research Question

How can the validity of a learned vehicle dynamics residual be estimated during operation and used to adapt an MPC/NMPC controller under operating conditions not represented during training?

1.6 Specific Research Questions

RQ1: Which types of physically relevant changes in vehicle dynamics cause a significant loss of validity of the learned residual?

RQ2: What information do epistemic uncertainty, aleatoric uncertainty, calibration, and an out-of-distribution (OOD) indicator provide about control-relevant model error?

RQ3: To what extent are uncertainty, calibration, and OOD signals complementary or redundant in characterizing validity during operation?

RQ4: How can a confidence/validity signal

$$C_k = g(U_{\text{epi}}, U_{\text{ale}}, \text{Cal}, \text{OOD})$$

be constructed to reflect the error relevant to the controller?

RQ5: How should C_k be used to adapt the MPC/NMPC horizon and reduce closed-loop performance degradation when the validity of the learned model decreases?

RQ6: What improvement does the proposed architecture offer over a controller based solely on the physical model, a controller using the learned model without monitoring its validity, and architectures using only validity or OOD information?

1.7 Research Hypothesis

A calibrated combination of epistemic uncertainty, aleatoric uncertainty, and evidence of out-of-distribution operation can provide a useful estimate of the validity of a learned vehicle dynamics residual during operation. Incorporating this estimate into the adaptation of an MPC/NMPC controller may reduce closed-loop perfor-

mance degradation under regime changes, compared with architectures that use the learned model without considering its validity.

The remainder of this document is organized as follows. Chapter 2 establishes the research objectives; Chapters 3 and 4 present the fundamentals of vehicle dynamics and control, together with the concepts needed to characterize the validity of learned information during operation. Chapter 5 reviews the state of the art and substantiates the research gap. Chapters 6 and 7 develop the proposed architecture and the methodology used for its evaluation. Chapters 8 and 9 present the preliminary results and expected contributions, respectively. Finally, Chapters 10–12 describe the work plan, the risks and limitations of the study, and the conclusions of the qualifying examination.

Chapter 2

Objectives

2.1 General Objective

To develop and evaluate a real-time validity estimation methodology for a hybrid vehicle dynamics model comprising a physical model and a learned residual, and to integrate this estimate into an MPC/NMPC controller to reduce closed-loop performance degradation under operating conditions not represented during training.

2.2 Specific Objectives

SO1: Formulate a hybrid vehicle dynamics architecture comprising a control-oriented physical model and a learned residual.

SO2: Operationally define the types of change and out-of-distribution situations relevant to the study, distinguishing at least unrepresented physical regimes and out-of-distribution dynamics, and including out-of-distribution inputs if time permits.

SO3: Estimate and calibrate the aleatoric and epistemic uncertainty of the learned model using a heteroscedastic model and a small ensemble, complemented by an OOD indicator with low computational cost.

SO4: Experimentally characterize the relationships among distribution shift, uncertainty, calibration, OOD, prediction error, and control-relevant error.

SO5: Define a runtime confidence/validity representation

$$C_k = g(U_{\text{epi}}, U_{\text{ale}}, \text{Cal}, \text{OOD})$$

and evaluate its ability to anticipate or represent model degradation in a way that is useful to the controller.

SO6: Integrate the validity signal into an MPC/NMPC controller through horizon adaptation and evaluate its effect on closed-loop performance and empirical robustness.

SO7: Compare the proposed approach with reference architectures that isolate the effects of the learned model, validity estimation, and out-of-distribution detection.

SO8: As an extension, if the schedule permits, evaluate the use of the validity signal in a CBF and/or a DAgger-based data acquisition and retraining strategy for conditions associated with epistemic uncertainty.

Chapter 3

Fundamentals of Vehicle Dynamics and Control

3.1 Vehicle Dynamics Relevant to Control

The lateral motion considered in this thesis is described by a linear dynamic bicycle model. This representation combines the wheels on each axle into a single equivalent wheel and assumes small slip angles, linear cornering stiffnesses, and constant longitudinal speed around an operating point. For path tracking, the model is augmented with lateral and heading errors, allowing the relevant states to be represented without reproducing the full multibody dynamics of the high-fidelity plant. The resulting state vector is

$$\mathbf{x} = \begin{bmatrix} e_y & e_\psi & v_y & r \end{bmatrix}^\mathsf{T}, \tag{3.1}$$

where $e_\psi = \psi - \psi_{\text{ref}}$. The control input is the effective front-wheel steering angle, δ, while the reference path curvature, κ_{ref}, is a known input.

Figure 3.1 illustrates the vehicle representation and the variables associated with tracking the reference path.

The model equations are

$$\dot{e}_y = V_{x,0} e_\psi + v_y, \tag{3.2}$$

$$\dot{e}_\psi = r - V_{x,0} \kappa_{\text{ref}}, \tag{3.3}$$

$$\dot{v}_y = -\frac{C_f + C_r}{m V_{x,0}} v_y + \left(\frac{C_r l_r - C_f l_f}{m V_{x,0}} - V_{x,0} \right) r + \frac{C_f}{m} \delta, \tag{3.4}$$

$$\dot{r} = \frac{C_r l_r - C_f l_f}{I_z V_{x,0}} v_y - \frac{C_f l_f^2 + C_r l_r^2}{I_z V_{x,0}} r + \frac{C_f l_f}{I_z} \delta. \tag{3.5}$$

The cornering stiffnesses C_f and C_r represent equivalent values per axle. Lateral forces are approximated linearly, without explicit grip saturation.

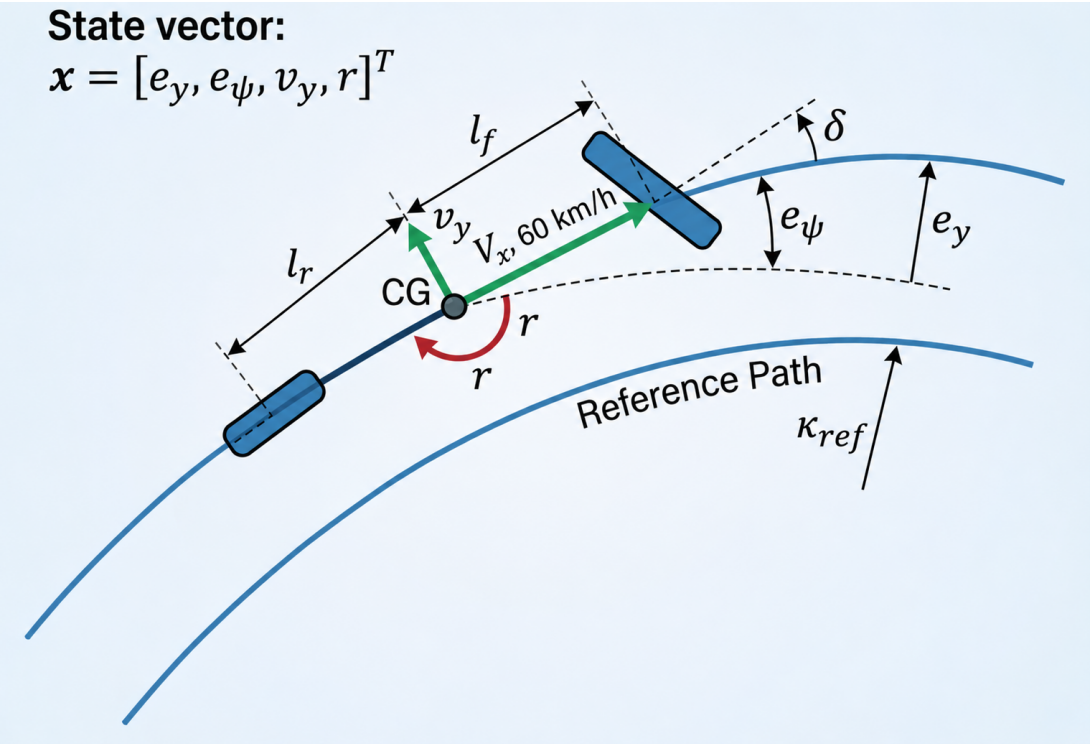


Figure 3.1: Schematic of the bicycle model and the variables used to describe lateral motion and path tracking.

Table 3.1: States and inputs of the augmented bicycle model.

Symbol	Description	Unit
e_y	Lateral path error	m
e_ψ	Heading error	rad
v_y	Lateral velocity at the center of gravity	m/s
r	Yaw rate	rad/s
δ	Effective front-wheel steering angle	rad
κ_{ref}	Reference path curvature	m^{-1}

Mass, yaw moment of inertia, center of gravity position, and cornering stiffnesses determine the lateral and yaw response. In particular, the stiffnesses C_f and C_r describe the local slope of the force–slip angle relationship and should not be confused with the force limit imposed by tire–road grip. When the vehicle operates near saturation, its load changes, or it moves away from the nominal speed operating point, the linear assumptions may no longer adequately represent the plant response.

3.2 Control-Oriented Models and Model Discrepancies

For control purposes, the preceding equations are organized as a state-space model. This formulation preserves the dominant relationships between steering, lateral motion, and yaw, and yields a discrete model compatible with the controller's update rate.

From these equations, the continuous state-space representation is

$$\dot{\mathbf{x}} = \mathbf{A}_c\mathbf{x} + \mathbf{B}_c\delta + \mathbf{E}_c\kappa_{\text{ref}}, \tag{3.6}$$

with

$$\mathbf{A}_c = \begin{bmatrix} 0 & V_{x,0} & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -\dfrac{C_f + C_r}{mV_{x,0}} & \dfrac{C_r l_r - C_f l_f}{mV_{x,0}} - V_{x,0} \\ 0 & 0 & \dfrac{C_r l_r - C_f l_f}{I_z V_{x,0}} & -\dfrac{C_f l_f^2 + C_r l_r^2}{I_z V_{x,0}} \end{bmatrix}, \tag{3.7}$$

$$\mathbf{B}_c = \begin{bmatrix} 0 \\ 0 \\ C_f/m \\ C_f l_f/I_z \end{bmatrix}, \qquad \mathbf{E}_c = \begin{bmatrix} 0 \\ -V_{x,0} \\ 0 \\ 0 \end{bmatrix}. \tag{3.8}$$

For use in MPC prediction, the model is discretized using a *zero-order hold* (ZOH), assuming constant inputs over each sampling interval:

$$\mathbf{x}_{k+1} = \mathbf{A}_d\mathbf{x}_k + \mathbf{B}_d\delta_k + \mathbf{E}_d\kappa_{\text{ref},k}, \tag{3.9}$$

where

$$\mathbf{A}_d = \exp\left(\mathbf{A}_c T_s\right), \tag{3.10}$$

$$\begin{bmatrix} \mathbf{B}_d & \mathbf{E}_d \end{bmatrix} = \int_0^{T_s} \exp\left(\mathbf{A}_c\tau\right) \begin{bmatrix} \mathbf{B}_c & \mathbf{E}_c \end{bmatrix} \mathrm{d}\tau. \tag{3.11}$$

The matrices are calculated at the nominal speed $V_{x,0}$ and remain constant throughout the simulation. This linear internal model, consisting solely of the bicycle model, is

denoted B0 and constitutes the physical basis into which the learned correction is incorporated.

The simplification that makes B0 computationally useful also causes discrepancies relative to CarSim. Sources include tire linearization, omitted saturation and coupling effects, errors in nominal parameters, and changes in operating regime. An unknown but constant actual parameter can produce model error without constituting a distribution shift; a distribution shift occurs when operation departs from the conditions represented by the data or the assumptions used to construct the model.

3.3 Hybrid Models: Physics and Learned Residuals

A hybrid model retains the physical prediction as its underlying structure and adds a learned function to approximate the discrepancy that this structure does not explain. In general form,

$$\mathbf{x}_{k+1} = f_{\text{phys}}(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{r}_{\boldsymbol{\theta}}(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{w}_k, \tag{3.12}$$

where f_{phys} is the nominal model, $\mathbf{r}_{\boldsymbol{\theta}}$ is the learned residual, and $\mathbf{w}_k$ groups disturbances and unrepresented discrepancies. In architecture B1, this idea is specialized to the lateral states of the bicycle model.

To complement the physical description of B0, a residual model is incorporated to estimate prediction discrepancies in lateral velocity and yaw rate. The combination of the bicycle model and this learned correction is denoted B1. Its purpose is to represent part of the dynamics not captured by the nominal model's approximations, while retaining the physical structure as the basis for prediction.

The lateral state vector is defined as

$$\boldsymbol{\xi}_k = \begin{bmatrix} v_{y,k} \\ r_k \end{bmatrix}. \tag{3.13}$$

The physical prediction of these states is obtained through

$$\hat{\boldsymbol{\xi}}^{\text{B0}}_{k+1|k} = \mathbf{A}_{\text{lat}}\boldsymbol{\xi}_k + \mathbf{B}_{\text{lat}}\delta_k, \tag{3.14}$$

where $\mathbf{A}_{\text{lat}} \in \mathbb{R}^{2\times 2}$ and $\mathbf{B}_{\text{lat}} \in \mathbb{R}^{2\times 1}$ are the submatrices of $\mathbf{A}_d$ and $\mathbf{B}_d$ corresponding to states v_y and r, respectively.

Once the physical prediction has been defined, the residual used as the training target is obtained as the difference between the CarSim response at the next instant and the one-step physical prediction:

$$\mathbf{d}_k = \boldsymbol{\xi}^{\text{CarSim}}_{k+1} - \hat{\boldsymbol{\xi}}^{\text{B0}}_{k+1|k} = \begin{bmatrix} d_{v_y,k} \\ d_{r,k} \end{bmatrix}. \tag{3.15}$$

Both responses are compared for the same transition, using the corresponding initial state and steering input. The prediction interval matches the controller's sampling period, T_s. Therefore, the residual components have units of m/s and rad/s: they represent discrete corrections to the states, not to their derivatives.

This residual is approximated using a *multilayer perceptron* (MLP) neural network, whose input vector is

$$\mathbf{z}_k = \begin{bmatrix} V_{x,k} & v_{y,k} & r_k & \delta_k \end{bmatrix}^\mathsf{T}. \tag{3.16}$$

The network provides a deterministic estimate of the two residual components:

$$\hat{\mathbf{d}}_k = \mathbf{g}_{\boldsymbol{\theta}}(\mathbf{z}_k), \tag{3.17}$$

where $\boldsymbol{\theta}$ groups its weights and biases. Although measured longitudinal speed is included among the network inputs, the physical model matrices remain fixed at the nominal operating point.

Once the residual has been trained, the lateral prediction of the hybrid model is constructed by adding the learned correction to the B0 prediction:

$$\boxed{\hat{\boldsymbol{\xi}}^{\mathrm{B1}}_{k+1|k} = \mathbf{A}_{\mathrm{lat}}\boldsymbol{\xi}_k + \mathbf{B}_{\mathrm{lat}}\delta_k + \mathbf{g}_{\boldsymbol{\theta}}(\mathbf{z}_k).} \tag{3.18}$$

The residual is added directly to the physical prediction, without multiplying it by T_s again, because it already represents a discrepancy over one sampling interval.

The validity of this correction depends on the domain covered by the data. Outside that domain, a deterministic residual may produce a numerically plausible correction without indicating that its error has increased. This limitation motivates estimating uncertainty, calibration, and distance from the training domain during operation, and relating these signals to the effect of error on control.

3.4 Model Predictive Control

MPC computes an input sequence using a model to predict system evolution over a finite horizon and explicitly account for its constraints [17]. For a prediction horizon N, a representative formulation is

$$\begin{aligned} \min_{\mathbf{u}_{0|k},\ldots,\mathbf{u}_{N-1|k}} \quad & \sum_{i=0}^{N-1} \left(\|\mathbf{x}_{i|k} - \mathbf{x}^{\mathrm{ref}}_{i|k}\|^2_{\mathbf{Q}} + \|\mathbf{u}_{i|k}\|^2_{\mathbf{R}} \right) + \|\mathbf{x}_{N|k} - \mathbf{x}^{\mathrm{ref}}_{N|k}\|^2_{\mathbf{P}}, \\ \text{subject to} \quad & \mathbf{x}_{i+1|k} = f_{\mathrm{pred}}(\mathbf{x}_{i|k}, \mathbf{u}_{i|k}), \\ & \mathbf{x}_{i|k} \in \mathcal{X}, \qquad \mathbf{u}_{i|k} \in \mathcal{U}. \end{aligned} \tag{3.19}$$

The matrices $\mathbf{Q}$, $\mathbf{R}$, and $\mathbf{P}$ weight tracking, control effort, and the terminal state; $\mathcal{X}$ and $\mathcal{U}$ represent the constraints. If f_{pred} is linear, the result is a linear MPC formulation; if it retains relevant nonlinearities, the problem corresponds to NMPC.

Only the first input of the optimized sequence is applied to the plant. At the next instant, the state is updated, the horizon is shifted, and the problem is solved again. This receding-horizon principle allows new information to be incorporated, but does not eliminate dependence on model quality: a persistent error in f_{pred} propagates along the horizon and may modify both the predicted cost and constraint activation.

In this thesis, the horizon N is the main candidate variable for adapting the controller based on the validity signal. The direction and magnitude of this adaptation are not assumed in advance: they must be determined by analyzing the trade-off between predictive reach, model error accumulation, closed-loop performance, and computational cost.

Chapter 4

Learned Information and Runtime Validity

4.1 Learned Information Relevant to the Controller

Information produced by a learned model is relevant to control when it enters the representation of dynamics, state estimation, or a controller decision. Learning can be used, for example, to represent forces or dynamics components, estimate states or parameters, and correct nominal model predictions. These functions are not equivalent: a learned force model introduces a specific description of a subsystem, an estimator provides a physical quantity or state, and a residual model directly approximates the prediction discrepancy without separating its causes [24, 2, 11].

In this thesis, the main learned information is the discrete residual defined in Equation (3.15). From the variables available in $\mathbf{z}_k$, the learned model estimates the correction $\hat{\mathbf{d}}_k$ to lateral velocity and yaw rate. This correction is incorporated into the transition used by the MPC/NMPC, according to Equation (3.18). Thus, the residual is neither a control action nor a direct estimate of grip, mass, or tire cornering stiffness: it represents the combined effect of structural simplifications, parametric errors, and other discrepancies on the one-step prediction. This formulation allows model error to be learned without separately identifying each physical cause, as in other residual learning approaches for predictive control [11, 7].

The architecture thus distinguishes three levels of information. Actual parameters belong to the plant and may change between scenarios; nominal parameters determine physical model B0; and measured or estimated signals form the inputs available for computing predictions. In the proposed protocol, V_x, v_y, r, and δ are available to the residual model, whereas grip and the parameters characterizing mass, inertias, or tires are not supplied as inputs when the experiment defines them as unknown. Consequently, a latent physical change may modify the model–plant discrepancy even when the observed

variables remain within familiar ranges.

The relevance of the residual is not determined solely by its instantaneous error. As part of the transition model, its corrections may propagate along the prediction horizon and modify the input sequence selected by the controller. In probabilistic configurations, the residual prediction is also accompanied by a characterization of its dispersion or model uncertainty. These additional signals do not replace the residual or constitute a validity measure on their own; their purpose is to provide evidence for assessing whether the learned correction remains adequate for control decisions. The following sections clarify the interpretation of uncertainty, distribution shifts, and runtime validity.

4.2 Predictive Uncertainty

A point prediction of the residual indicates the correction that the model considers most likely, but does not express how much dispersion it assigns to that estimate. Probabilistic models represent each component $d_{j,k}$ through a predictive distribution conditioned on the input and the training data,

$$p(d_{j,k} \mid \mathbf{z}_k, \mathcal{D}_{\mathrm{tr}}), \qquad j \in \{v_y, r\}. \tag{4.1}$$

In the configurations studied, this distribution is described by a mean $\mu_{j,k}$ and a predictive variance. To facilitate analysis, the total variance is operationally decomposed as

$$U^2_{\mathrm{tot},j,k} = U^2_{\mathrm{ale},j,k} + U^2_{\mathrm{epi},j,k}, \tag{4.2}$$

where U_{ale} represents the conditional dispersion attributed to random variability and U_{epi} the uncertainty associated with model knowledge. This separation depends on the model and estimation procedure: its components are not directly observed physical quantities, nor are they uniquely identified from the data.

Predictive uncertainty is used here as supervisory information for the residual, rather than as a synonym for error. A high standard deviation does not demonstrate that the mean is incorrect, and a small deviation does not guarantee that the prediction is valid. Its usefulness must be established by comparing it with observed errors, distribution shifts, and, ultimately, the consequences of those errors for the controller.

4.2.1 Aleatoric Uncertainty

Aleatoric uncertainty describes the variability remaining in the output after conditioning the prediction on the variables supplied to the model. It may encompass measurement noise, unobserved disturbances, and plant variations that the selected inputs cannot explain. It is considered irreducible under the available representation and information,

although it may change if new variables are incorporated, the model is modified, or the measurement system improves. For this reason, the separation between aleatoric uncertainty and modeling error must be interpreted relative to the adopted assumptions, rather than as an absolute classification [5].

When constant dispersion is assumed throughout the input space, the model is homoscedastic. This assumption is used in B2, where the term $\sigma_{n,j}^2$ in Equation (7.6) is a single value for each residual component. A heteroscedastic model instead allows dispersion to depend on $\mathbf{z}_k$:

$$d_{j,k} = \mu_j(\mathbf{z}_k) + \varepsilon_{j,k}, \qquad \varepsilon_{j,k} \sim \mathcal{N}\big(0, U_{\text{ale},j}^2(\mathbf{z}_k)\big)\,. \tag{4.3}$$

B3 learns this dispersion together with the mean using a heteroscedastic Gaussian loss function, whereas B4 estimates it in a second stage from out-of-fold residuals. In both cases, U_{ale} should be read as an operational estimate of residual dispersion. It may contain not only physical noise, but also mean-model discrepancies and unexplained variability between runs.

A region with high conditional dispersion may still belong to the training domain, and an unobserved physical change may produce systematic error without increasing this component. Consequently, aleatoric uncertainty provides information about the expected magnitude of errors, but does not by itself establish that the residual has lost validity.

4.2.2 Epistemic Uncertainty

Epistemic uncertainty represents the lack of knowledge associated with finite data, insufficient domain coverage, and ambiguity in model fitting. Unlike the aleatoric component, it can in principle be reduced through informative observations in the relevant regions, although the actual reduction depends on the ability of the model family to represent the dynamics [5].

This component is approximated in two ways in the probabilistic configurations. Gaussian processes B2 and B4 use the posterior standard deviation of the latent function, which depends on the location of the query relative to the training data and on the selected kernel. B3 uses disagreement among ensemble member means:

$$U_{\text{epi},j,k}^2 = \frac{1}{M}\sum_{m=1}^{M}\Big(\mu_{j,k}^{(m)} - \bar{\mu}_{j,k}\Big)^2, \qquad \bar{\mu}_{j,k} = \frac{1}{M}\sum_{m=1}^{M}\mu_{j,k}^{(m)}. \tag{4.4}$$

Disagreement reflects sensitivity to different solutions obtained during training, but does not cover every source of missing knowledge. Members may share data, architecture, and representation biases and therefore agree on an erroneous prediction.

U_{epi} is expected to increase in sparsely represented regions, but this response is not

guaranteed for every change. In particular, a change in grip, mass, or tires not included in $\mathbf{z}_k$ may alter the input–output relationship without visibly shifting the inputs. Epistemic uncertainty and out-of-distribution detection are therefore related but distinct concepts: a distant input may not cause a relevant error, and a latent dynamic change may produce a confident failure.

4.2.3 Calibration

Calibration compares predictive distributions with subsequently observed errors. For a Gaussian approximation with standard deviation $U_{\mathrm{tot},j,k}$, a two-sided interval with nominal level $1-\alpha$ is defined as

$$\mathcal{I}_{j,k}^{1-\alpha} = \left[\mu_{j,k} - q_{1-\alpha/2} U_{\mathrm{tot},j,k},\, \mu_{j,k} + q_{1-\alpha/2} U_{\mathrm{tot},j,k}\right], \tag{4.5}$$

where $q_{1-\alpha/2}$ is the corresponding standard normal quantile. Empirical coverage over N observations is

$$\widehat{\mathrm{Cov}}_j^{\,1-\alpha} = \frac{1}{N} \sum_{k=1}^{N} \mathbb{I}\left\{ d_{j,k} \in \mathcal{I}_{j,k}^{1-\alpha} \right\}. \tag{4.6}$$

Coverage below the nominal level indicates overconfidence, whereas higher coverage may indicate conservative intervals. However, coverage alone does not reward sharpness: unnecessarily wide intervals may cover many observations while providing little information.

For this reason, the Gaussian negative log-likelihood is also used, combining mean error with the declared dispersion:

$$\mathrm{NLL}_j = \frac{1}{N} \sum_{k=1}^{N} \left[\frac{1}{2} \log\left(2\pi U_{\mathrm{tot},j,k}^2\right) + \frac{\left(d_{j,k} - \mu_{j,k}\right)^2}{2 U_{\mathrm{tot},j,k}^2} \right]. \tag{4.7}$$

Joint evaluation of coverage and NLL can reveal both overconfidence and excessive dispersion. When the prediction comes from a mixture, as in ensemble B3, it must be stated whether the NLL corresponds to the exact mixture or to a moment-matched Gaussian approximation.

Calibration must be checked on partitions not used to fit the model and broken down by region and family of change. Adequate average coverage within the nominal domain may hide local errors and does not guarantee that intervals remain calibrated when the dynamics change. Furthermore, calibrating one-step residual error does not demonstrate that the uncertainty represents accumulated error over the horizon or its closed-loop effect. Calibration therefore provides a quality condition for interpreting probabilistic signals, but is not a sufficient definition of validity for control.

4.3 Distribution Shifts and OOD Detection in Vehicle Dynamics

The training domain is not defined solely by the ranges of individual input variables. It also encompasses combinations of states and inputs, the maneuvers generating those combinations, and the dynamic relationship between model inputs and the observed residual. To express this distinction, the joint distribution can be factorized as

$$p(\mathbf{z}, \mathbf{d}) = p(\mathbf{d} \mid \mathbf{z})\, p(\mathbf{z}), \tag{4.8}$$

where $\mathbf{z}$ contains the variables available to the predictor and $\mathbf{d}$ is the model–plant discrepancy. During operation, the input distribution $p(\mathbf{z})$, the conditional relationship $p(\mathbf{d} \mid \mathbf{z})$, or both may change. This separation matters because a detector based exclusively on inputs can identify departure from the observed domain, but not necessarily a dynamics change conditioned on a parameter that the model does not receive.

In this work, the term out of distribution is always used relative to an explicit reference domain and a specific representation. It is not considered an absolute property of a sample. Furthermore, detecting a distribution shift does not demonstrate that the residual has lost usefulness, just as a large error does not require statistically atypical inputs. The OOD indicator is intended to provide a departure signal that must be compared with predictive error and its effect on control. This caution is especially relevant in learning-based control, where distribution shifts and data scarcity may affect generalization without independently determining the closed-loop consequence [5].

4.3.1 Unrepresented Physical Regime

A physical regime is characterized by tire–road contact conditions and the parameters determining the vehicle response. In compact form, these quantities can be grouped as

$$\boldsymbol{\vartheta}_k = \begin{bmatrix} \mu_k & m_k & I_{z,k} & C_{f,k} & C_{r,k} & \cdots \end{bmatrix}^\mathsf{T}, \tag{4.9}$$

where μ represents grip, m mass, I_z yaw moment of inertia, and C_f and C_r equivalent cornering stiffnesses. An unrepresented physical regime occurs when the configuration used during operation does not belong to the set of configurations used to generate the training data. The definition therefore depends on which parameters were varied and which ranges were covered in that set.

An unknown parameter must be distinguished from an unrepresented parameter. If a physical value remains fixed during both training and operation, the residual may learn part of the discrepancy it produces even if the controller does not know that value. There is parametric uncertainty relative to the physical model, but not necessarily a distribution

shift. Conversely, if grip, mass, load, or tire properties depart from the configurations used to train the residual, the relationship between $\mathbf{z}_k$ and $\mathbf{d}_k$ may change. Studies on adaptive control and modeling show that these quantities play different roles in vehicle response and that their variation may require adapting the model or controller [21, 12, 24].

In this thesis architecture, the actual values of $\boldsymbol{\vartheta}_k$ are used to configure the reference plant and describe experimental scenarios, but are not supplied to the residual when the protocol declares them unavailable. Thus, a reduction in grip or a change of vehicle may constitute a latent physical change: the predictor receives V_x, v_y, r, and δ, but no label directly identifying the new configuration. The objective is not to recover each component of $\boldsymbol{\vartheta}_k$, but to determine whether the learned correction continues to describe the discrepancy usefully for the controller.

Not every physical change implies a loss of validity. A moderate variation may have a small effect on the corrected states or be partially accommodated by residual generalization. Scenarios must therefore be organized by family and severity and evaluated against a nominal reference. The unrepresented-regime label identifies the experimental cause of change; residual degradation and its control relevance must be measured separately.

4.3.2 Out-of-Distribution Dynamics

Out-of-distribution dynamics refers to a transition or operating region whose relationship between state, input, and response is not sufficiently represented by the training data. In terms of Equation (4.8), the most direct case occurs when

$$p_{\mathrm{op}}(\mathbf{d} \mid \mathbf{z}) \neq p_{\mathrm{tr}}(\mathbf{d} \mid \mathbf{z}), \tag{4.10}$$

for a region relevant to the maneuver. The inequality represents an operational difference to be quantified through error and the selected indicators; it does not presume that the distributions are known analytically.

This situation can arise in two ways. In the first, the vehicle enters an observable region of the operating space not covered during training. For example, if the residual was fitted at a single longitudinal speed, operating below or above it requires extrapolation with respect to V_x, even if the physical configuration remains fixed. In the second, inputs may retain familiar values while a latent plant variable changes. A change in grip, mass, or tires may then produce a different residual for similar values of $\mathbf{z}_k$. The first case combines covariate shift with possible dynamic extrapolation; the second corresponds to a conditional shift that need not be visible in input space.

This distinction determines what each mechanism can detect. A distance or density indicator computed on $\mathbf{z}_k$ is suitable for flagging poorly covered state and input regions, but may remain low under a latent physical change. An indicator based on innovations or observed errors may respond to the latter, although only after the plant response becomes

available, and may confuse dynamic change with noise or a sensor fault. Therefore, no single indicator is adopted as universal proof of OOD dynamics.

The methodology implements these categories through separate scenarios. Speed changes represent observable extrapolations of the operating point; variations in grip and vehicle configuration represent latent dynamic changes; and changes during a run allow the temporal response of the indicators to be studied. Tables 7.21 and 7.22 document the nominal domain, available information, and planned comparison for each case. This classification allows whether a change was detected, whether it degraded prediction, and whether that degradation was relevant to control to be analyzed as related but distinct questions.

4.3.3 Out-of-Distribution Inputs

4.3.4 Sensor Failure or Degradation

4.4 Runtime Validity and Confidence

In this thesis, residual *validity* is understood as its adequacy for the role it plays within the controller's prediction model under current conditions. A residual is operationally valid when its correction maintains sufficient quality to support MPC/NMPC decisions. This definition is relative to the task, horizon, and variables affecting control; it does not require the learned model to identify the physical cause of the discrepancy or reproduce every aspect of the plant with equal accuracy.

Four concepts should be distinguished. Predictive uncertainty expresses the dispersion the model assigns to its prediction; the OOD indicator characterizes departure from a reference domain; validity describes the residual's actual adequacy for its intended use; and confidence is an estimate of that validity available during operation, before the future plant response is known. In compact form,

$$\text{uncertainty} \neq \text{OOD} \neq \text{validity} \neq \text{confidence}. \tag{4.11}$$

The four notions may be related, but none automatically substitutes for the others. A point may lie far from the data and still admit a useful correction; a large, confident error may also occur under a latent dynamic change.

Runtime confidence is represented by a scalar signal C_k, normalized so that larger values indicate greater support for using the residual. The working formulation is

$$C_k = g(U_{\text{epi},k}, U_{\text{ale},k}, \text{Cal}_k, S_{\text{OOD},k}), \qquad 0 \leq C_k \leq 1, \tag{4.12}$$

where $S_{\text{OOD},k}$ is the selected departure score and Cal_k represents the calibration informa-

tion or transformation applicable to the probabilistic signals. The latter need not be a new instantaneous measurement: it may correspond to recalibration parameters obtained from a held-out set and, if the method allows, a local diagnostic. The form of g is not fixed a priori. It must be determined after studying the sensitivity, redundancy, and complementarity of its components; if a signal adds no information, the final representation may omit it.

C_k is not interpreted as the probability that the prediction is correct unless a validity event is defined and that probabilistic calibration is demonstrated empirically. In its initial formulation, it is an index of relative support. Nor does it estimate μ, m, C_f, or C_r: it should respond to degradation without identifying which of those parameters changed. This separation allows the same mechanism to be evaluated across different families of change, although its thresholds and sensitivity may not be universal.

During operation, $U_{\mathrm{epi},k}$, $U_{\mathrm{ale},k}$, and $S_{\mathrm{OOD},k}$ can be computed from the current input before observing $\mathbf{d}_k$. Actual validity can only be assessed subsequently, when the plant transition or a closed-loop performance measure becomes available. The design of C_k must therefore separate the data used to train the residual, calibrate the signals, select function g, and evaluate generalization. Without this separation, the index might describe the scenarios used to tune it without reliably estimating validity during new operation.

4.5 Control-Relevant Validity

One-step residual accuracy constitutes the first level of evaluation, but does not by itself determine control relevance. If $\mathbf{d}_k$ is the actual discrepancy relative to B0 and $\hat{\mathbf{d}}_k$ the learned correction, the physical and hybrid model errors can be written as

$$\mathbf{e}_{d,k}^{\mathrm{B0}} = \mathbf{d}_k, \qquad \mathbf{e}_{d,k}^{\mathrm{hyb}} = \mathbf{d}_k - \hat{\mathbf{d}}_k. \tag{4.13}$$

For a loss function ℓ_d, the instantaneous predictive gain of the residual over the physical model is

$$G_k^{(1)} = \ell_d\big(\mathbf{e}_{d,k}^{\mathrm{B0}}\big) - \ell_d\Big(\mathbf{e}_{d,k}^{\mathrm{hyb}}\Big). \tag{4.14}$$

A positive value indicates that the correction improves one-step prediction; a negative value indicates that it is harmful for that transition. This measure allows checking whether learning adds value relative to B0, but does not yet capture error propagation or optimization sensitivity.

The second level is accumulated dynamic error. Applying the model recursively over a horizon H yields

$$\mathbf{e}_{x,i|k} = \mathbf{x}_{k+i} - \hat{\mathbf{x}}_{k+i|k}, \qquad i = 1, \ldots, H. \tag{4.15}$$

Even a small one-step bias may accumulate, whereas errors of alternating sign may par-

tially cancel. Furthermore, the controller itself modifies future inputs, so evaluating open-loop trajectories does not necessarily reproduce the state distribution visited in closed loop.

The third level is decision relevance. The MPC/NMPC weights states, inputs, and constraints through the cost function in Equation (3.19). Consequently, errors of equal statistical magnitude may have different effects: a discrepancy in a low-sensitivity direction may leave the first optimized action unchanged, while a smaller error near a constraint or during a demanding maneuver may alter the control sequence. A control-relevant error measure must therefore account for propagation over the horizon, problem weights, and proximity to constraints, rather than only the predictor's average RMSE.

The fourth level is observed closed-loop performance. Estimated validity is useful to this thesis only if using it reduces tracking degradation, constraint violations, or control cost under the changes studied while maintaining computational feasibility. The relationship to be evaluated is therefore

$$\begin{aligned} &\text{condition change} \longrightarrow \text{residual error} \\ &\qquad\qquad \longrightarrow \text{effect on optimization} \longrightarrow \text{closed-loop performance.} \end{aligned} \tag{4.16}$$

This chain avoids defining validity solely through an OOD label or a probabilistic metric.

Evaluation of C_k must check its association with all four levels: predictive gain over B0, multistep error, relevant changes in decisions, and closed-loop degradation. It must also include cases where a physical change does not harm the residual and cases of confident failure. This avoids rewarding a signal that responds to every change even when it does not matter to the controller. The specific MPC/NMPC adaptation is defined in the methodological chapters; conceptually, C_k informs how much to trust the correction, but does not constitute a formal guarantee of stability, recursive feasibility, or safety.

Chapter 5

State of the Art

5.1 Literature Review Strategy

5.2 Learned and Hybrid Models for Vehicle Dynamics Control

5.3 Uncertainty-Aware Learned Dynamics for Control

5.4 Out-of-Distribution Detection, Distribution Shifts, and Runtime Monitoring

5.5 Validity- and Confidence-Aware Control

5.6 Critical Synthesis and Research Gap

Chapter 6

Conceptual Framework and Proposed Architecture

6.1 The Shift–Validity–Control Chain

The proposed architecture is organized around a causal chain linking operating changes to their possible consequences for the controller. The starting point is a modification of the physical regime, operating range, or excitation relative to the conditions used to train the residual. Change may occur between trips, such as a different load, or during a run, such as a reduction in grip. Its actual values are used to define the plant and evaluation scenarios, but are not provided to the supervisor when the protocol declares them unknown.

Figure 6.1 summarizes this sequence and the feedback used during operation. The

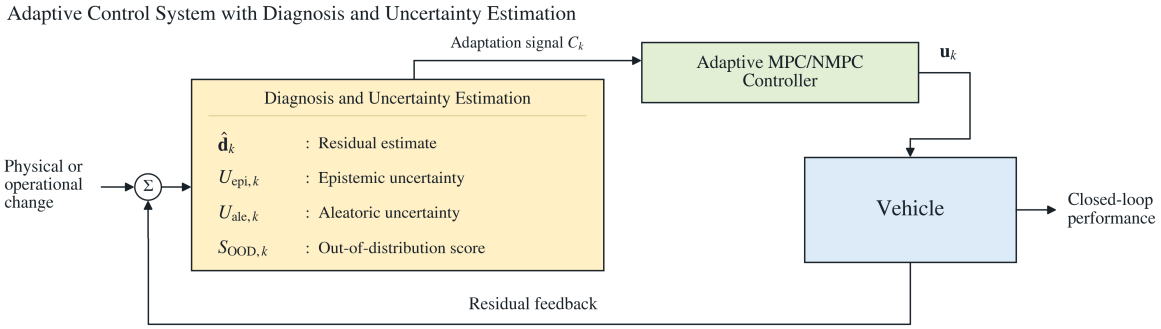


Figure 6.1: Shift–validity–control chain and residual feedback in the proposed architecture.

first arrow does not imply that every change invalidates the learned correction. It indicates that the change may modify the discrepancy between B0 and the plant. Likewise, uncertainty and OOD signals do not constitute validity: they are candidate observations for constructing the confidence C_k defined in Equation (4.12). Each link in the chain must

be checked independently.

At time k, the controller has the measured or estimated signals forming $\mathbf{z}_k$. The learned block computes the correction $\hat{\mathbf{d}}_k$ and the available uncertainty components, while the OOD monitor characterizes departure from the training domain. The validity estimator transforms these signals into C_k without necessarily receiving grip, mass, inertias, or tire parameters. Thus, the supervisor seeks to determine whether the correction retains sufficient support, rather than which physical parameter caused the change.

The signal C_k modifies a controller configuration represented by $\boldsymbol{\eta}_k$:

$$\boldsymbol{\eta}_k = h(C_k), \qquad \mathbf{u}_k = \kappa_{\mathrm{MPC}}(\hat{\mathbf{x}}_k, \mathbf{r}_k, f_{\mathrm{hyb}}, \boldsymbol{\eta}_k)\,, \tag{6.1}$$

where $\mathbf{r}_k$ represents the reference and f_{hyb} the hybrid prediction model. The main mechanism to be investigated is horizon adaptation, so that $\boldsymbol{\eta}_k = N_k$ in the initial formulation. Function h, its bounds, and the appropriate direction of adaptation are not assumed in advance: they must be determined through the relationship between horizon, error accumulation, computational cost, and closed-loop performance.

After $\mathbf{u}_k$ is applied, the plant evolves and generates the signals used in the next cycle. These observations allow retrospective evaluation of residual error and the effect of the decision, but the action at time k must be selected before they are known. This temporal separation is why C_k is formulated as a confidence estimate rather than a direct measure of future error. In the thesis core, the learned model parameters remain frozen throughout the campaign; updating or retraining constitutes an optional extension.

Validation traverses the chain in both directions. First, it checks whether the selected changes alter the error and whether the available signals reflect that alteration. It then examines whether C_k relates to control-relevant error and whether modifying the MPC/NMPC reduces degradation relative to architectures that do not monitor validity. A favorable result at one stage does not guarantee the next: detecting a change does not ensure it is harmful, estimating degradation does not ensure horizon adaptation is the appropriate response, and improving a predictive metric does not by itself demonstrate a closed-loop improvement. This structure makes each link a testable experimental hypothesis and defines the empirical scope of the conclusions.

6.2 Hybrid Prediction Model

The derivation of physical model B0 and its discretization is presented in Section 3.2; the definition of the residual target, its input variables, and the B1 prediction is given in Section 3.3. On that basis, this section does not reformulate the individual models, but establishes the interface through which any of configurations B1–B4 is incorporated into the MPC/NMPC predictor.

Let f_{B0} be the nominal transition of the complete state $\mathbf{x} = [e_y, e_\psi, v_y, r]^\mathsf{T}$, and let $\bar{\mathbf{d}}^{(b)}(\mathbf{z})$ be the mean correction provided by configuration b. Since the residual acts only on v_y and r, its insertion into the complete state is expressed through

$$\mathbf{J}_d = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad f_{\mathrm{hyb}}^{(b)}(\mathbf{x}, \delta, \kappa_{\mathrm{ref}}, \mathbf{z}) = f_{\mathrm{B0}}(\mathbf{x}, \delta, \kappa_{\mathrm{ref}}) + \mathbf{J}_d \bar{\mathbf{d}}^{(b)}(\mathbf{z}). \tag{6.2}$$

This notation retains the evolution of tracking errors in B0 and adds the learned correction only to the two variables for which it was trained. It also allows the purely physical reference to be represented within the same interface through $\bar{\mathbf{d}}^{(0)}(\mathbf{z}) = \mathbf{0}$; for $b \in \{1, 2, 3, 4\}$, the correction corresponds to the point prediction or mean of the residual distribution of the model considered.

At each stage i of the horizon, the learned block inputs are formed from the predicted variables available to the model:

$$\mathbf{z}_{i|k} = \begin{bmatrix} V_{x,i|k} & \hat{v}_{y,i|k} & \hat{r}_{i|k} & \delta_{i|k} \end{bmatrix}^\mathsf{T}, \qquad i = 0, \ldots, N_k - 1. \tag{6.3}$$

Here, $V_{x,i|k}$ denotes the value available for prediction, either through a known reference or through the convention of holding the current measurement over the horizon. The specific choice must be fixed in the control protocol and kept identical when comparing B0–B4. Starting from the estimated initial state, the recursive rollout is defined by

$$\hat{\mathbf{x}}_{0|k} = \hat{\mathbf{x}}_k, \qquad \hat{\mathbf{x}}_{i+1|k} = f_{\mathrm{hyb}}^{(b)}\left(\hat{\mathbf{x}}_{i|k}, \delta_{i|k}, \kappa_{\mathrm{ref},i|k}, \mathbf{z}_{i|k}\right). \tag{6.4}$$

Thus, although the residual is trained for one-step prediction, the controller evaluates it recursively at points generated by its own predicted trajectory. The distinction between one-step error, its multistep accumulation, and its effect on the decision is developed in Section 4.5.

The learned output is functionally divided into two paths. The residual mean $\bar{\mathbf{d}}^{(b)}$ modifies the transition in Equation (6.2); the available uncertainty quantities are neither added to the state nor substituted for the correction. They are sent, together with the OOD indicator and calibration information, to the validity estimator. The interpretation of these quantities is established in Sections 4.2 and 4.4, while their position within the complete architecture is shown in Section 6.1. Thus, the MPC receives corrected dynamics for optimization and, through a separate path, a signal that may modify its configuration without confusing uncertainty with dynamics.

The boundary between available information and hidden physical parameters is established in Section 4.3.1. Applied to Equation (6.2), it means that f_{B0} uses nominal

parameters and the learned block consumes only $\mathbf{z}_{i|k}$. Actual plant values are reserved for constructing scenarios and evaluating results; they are not part of the predictor interface. Thus, the comparison measures the contribution of the learned correction without providing the controller with privileged information about the physical change.

6.3 Validity Estimator

6.4 Integration with MPC/NMPC

Chapter 7

Research and Validation Methodology (Priority 1)

7.1 General Strategy

S1: Establish the control-oriented physical model and the reference MPC/NMPC controller.

S2: Train the learned residual and quantify its improvement and limitations relative to the physical model within the nominal domain.

S3: Implement aleatoric and epistemic uncertainty estimation, calibration, and an OOD score with low computational cost.

S4: Design physically interpretable changes in vehicle dynamics and characterize their effects on prediction, uncertainty, OOD, and validity.

S5: Construct and validate signal C_k against control-relevant error.

S6: Integrate C_k into MPC/NMPC adaptation.

S7: Compare the reference architectures and proposed approach under nominal and out-of-distribution conditions.

S8: Evaluate empirical robustness and, if progress permits, incorporate the CBF and/or DAgger extensions.

7.2 Common Model Training and Evaluation Environment

Models B1–B4 are built upon the same physical model B0 and use a common experimental environment. This decision allows observed differences to be attributed to the type of

residual model and its uncertainty representation, without confounding them with changes in data, partitioning, or preprocessing. This section documents the nominal model, identification domain, frozen split, and evaluation protocol once. The sections devoted to each model describe only their particular features.

7.2.1 Nominal Physical Model B0

Table 7.1: Parameters used in the nominal bicycle model B0.

Symbol	Description	Value	Unit
m	Vehicle mass	1501	kg
I_z	Yaw moment of inertia	2192.089539	$\mathrm{kg\,m^2}$
l_f	Distance from CG to front axle	1.049562958	m
l_r	Distance from CG to rear axle	1.860437042	m
C_f	Equivalent front-axle cornering stiffness	159055	N/rad
C_r	Equivalent rear-axle cornering stiffness	92776.5	N/rad
$V_{x,0}$	Nominal longitudinal speed	60/3.6	m/s
T_s	Sampling period	0.020	s

The discrete B0 matrices are obtained using a zero-order hold with the sampling period indicated in Table 7.1. They are calculated at the nominal speed operating point and kept fixed during baseline generation and evaluation. Actual parameters modified in the shift scenarios are recorded separately and are not provided to the controller unless explicitly stated in an experiment's protocol.

7.2.2 Data Generation and Nominal Domain

The data come from closed-loop simulations using the CarSim plant, the *C-Class hatchback* vehicle, and an MPC based exclusively on B0. The identification domain comprises *double lane change* (DLC) and *double lane change truncated* (DLCT) maneuvers, a reference speed of 60 km/h, and a grip coefficient $\mu = 1$. Signals are sampled at 50 Hz, so each target represents the discrepancy accumulated over an interval of 20 ms.

The dataset contains five complete runs of each maneuver. No samples were excluded from the nominal domain: the ten runs provide 5170 transitions, distributed as summarized in Table 7.2.

Table 7.2: Composition of the nominal dataset shared by B1–B4.

Maneuver	Runs	Samples per run	Total
DLC	001–005	634	3170
DLCT	001–005	400	2000
Total	10	—	5170

For all models, the input is the vector $\mathbf{z}_k = [V_{x,k},\, v_{y,k},\, r_k,\, \delta_k]^\mathsf{T}$ defined in Equation (3.16), and the target is the discrete residual $\mathbf{d}_k = [d_{v_y,k},\, d_{r,k}]^\mathsf{T}$ from Equation (3.15). Thus, the input matrix has dimensions 5170×4 and the target matrix 5170×2. The consistency of the stored residuals was verified by reconstructing both components from the B0 prediction and CarSim response; the maximum error in that check was numerically zero.

7.2.3 Frozen Split by Run

The split is performed by complete runs and remains unchanged for B1–B4. Thus, no transition from a run appears in more than one partition, and all comparisons use exactly the same data. For brevity, DLC-002 denotes run 002 of maneuver DLC, and similarly for the other groups.

Table 7.3: Shared frozen split of the nominal runs.

Partition	Runs	Samples
Training	DLC-001, DLCT-001, DLCT-002, DLC-004, DLCT-004 and DLC-005	3102
Validation	DLC-003 and DLCT-005	1034
ID test	DLC-002 and DLCT-003	1034

The validation partition is used to select configurations or stop training, depending on the algorithm. The ID test partition is excluded from those decisions and used only for the final comparison within the nominal domain. This explicit definition replaces a description based solely on a random seed and provides the reproducible reference for subsequent models.

7.2.4 Preprocessing and Comparison Protocol

Inputs and targets are standardized using statistics calculated exclusively from the 3102 training samples:

$$\tilde{\mathbf{z}}_k = \mathbf{S}_z^{-1}\left(\mathbf{z}_k - \boldsymbol{\mu}_z\right), \tag{7.1}$$

$$\tilde{\mathbf{d}}_k = \mathbf{S}_d^{-1}\left(\mathbf{d}_k - \boldsymbol{\mu}_d\right). \tag{7.2}$$

Vectors $\boldsymbol{\mu}_z$ and $\boldsymbol{\mu}_d$ contain the training means, while $\mathbf{S}_z$ and $\mathbf{S}_d$ are diagonal matrices containing the corresponding standard deviations. The same transformations are applied to validation and ID test data without refitting. A normalized prediction is returned to physical units through

$$\hat{\mathbf{d}}_k = \boldsymbol{\mu}_d + \mathbf{S}_d \hat{\tilde{\mathbf{d}}}_k. \tag{7.3}$$

Predictive ability is evaluated for each residual component using RMSE and MAE. In addition, the one-step hybrid prediction is compared with B0 on the same ID test transitions. Uncertainty and calibration metrics are added for probabilistic models, but are defined in the corresponding model-specific sections. Scenarios outside the nominal domain are not involved in training or selection and are reserved for studying loss of validity.

7.3 Description of the Trained Models

Each model is presented in terms of its role in the study, specific structure, training or selection criterion, and verification on the common partition. The data domain, variables, normalization, and split are not repeated in the individual descriptions.

7.3.1 B1: Deterministic Residual MLP

B1 is the deterministic residual baseline. Its role is to establish how much the B0 error can be reduced through a neural point correction, without yet incorporating a predictive distribution, uncertainty decomposition, or validity monitoring mechanism. B1 outputs an estimate of the two one-step residual components.

The MLP receives four standardized inputs, contains two hidden layers of 64 neurons with ReLU activation, and produces two linear outputs. In compact form,

$$\hat{\tilde{\mathbf{d}}}_k = \mathbf{W}_3\rho(\mathbf{W}_2\rho(\mathbf{W}_1\tilde{\mathbf{z}}_k + \mathbf{b}_1) + \mathbf{b}_2) + \mathbf{b}_3, \tag{7.4}$$

where $\rho(a) = \max(0, a)$ is applied componentwise. The output is destandardized through Equation (7.3) and added directly to the physical prediction, according to Equation (3.18).

Fitting is performed with MATLAB's `fitrnet` function. Limits of 100 and 200 iterations are compared, and the candidate with the lowest joint validation RMSE in standardized space is selected:

$$J_{\text{val}} = \sqrt{\frac{1}{2N_{\text{val}}} \sum_{k\in\mathcal{I}_{\text{val}}} \left\| \tilde{\mathbf{d}}_k - \hat{\tilde{\mathbf{d}}}_k \right\|_2^2}. \tag{7.5}$$

The 200-iteration limit achieved $J_{\text{val}} = 0.073810$, compared with 0.075873 for 100 iterations, and was the retained configuration. Table 7.4 summarizes only the decisions specific to B1.

Table 7.5 presents residual performance in all three partitions. The proximity of training and validation results indicates that selection does not depend on error reduction exclusive to the fitting samples; the ID test remains an independent check.

Finally, Table 7.6 compares the one-step B0 and B1 predictions on the ID test. B1

Table 7.4: Model B1 configuration.

Item	Configuration
Role in the study	Deterministic residual baseline
Model type	Regression MLP trained with `fitrnet`
Architecture	$4 \to 64 \to 64 \to 2$
Activation	ReLU in hidden layers and linear output
Output	Point estimates of $d_{v_y,k}$ and $d_{r,k}$
Evaluated limits	100 and 200 iterations
Selected configuration	200 iterations
Selection criterion	Lowest joint standardized validation RMSE
Predictive uncertainty	Unavailable

Table 7.5: B1 residual prediction performance at 20 ms.

	d_{v_y} (m/s)		d_r (rad/s)	
Partition	RMSE	MAE	RMSE	MAE
Training	0.009128	0.006627	0.007390	0.005420
Validation	0.008871	0.006100	0.007033	0.004664
ID test	0.010784	0.008180	0.008546	0.006490

reduces the RMSE of v_y by 91.66 % and that of r by 90.25 %. These results justify its use as a predictive baseline, but provide no information about correction validity outside the nominal domain.

Table 7.6: One-step comparison between B0 and B1 on the ID test.

State	RMSE B0	RMSE B1	Improvement
v_y (m/s)	0.129336	0.010784	91.66 %
r (rad/s)	0.087671	0.008546	90.25 %

7.3.2 B2: Homoscedastic Gaussian Process

B2 replaces the deterministic MLP in B1 with two independent Gaussian processes (GPs), one for d_{v_y} and one for d_r. Its role in the study is to retain correction ability comparable to B1 while adding a predictive distribution that quantifies uncertainty within and outside the nominal domain. Independence between the two outputs simplifies training, but means that B2 does not explicitly model their cross-covariance.

For each component $j \in \{v_y, r\}$, consider

$$d_{j,k} = f_j(\tilde{\mathbf{z}}_k) + \varepsilon_{j,k}, \qquad \varepsilon_{j,k} \sim \mathcal{N}(0, \sigma_{n,j}^2), \tag{7.6}$$

where f_j follows an exact GP with a squared-exponential kernel and automatic relevance

determination (ARD). For standardized inputs, the kernel can be written as

$$k_j(\mathbf{z}, \mathbf{z}') = \sigma_{f,j}^2 \exp\left[-\frac{1}{2}\sum_{q=1}^{4}\frac{(z_q - z_q')^2}{\ell_{j,q}^2}\right], \tag{7.7}$$

with one length scale $\ell_{j,q}$ per input. The term $\sigma_{n,j}$ is constant throughout the input space; this assumption defines the homoscedastic nature of B2.

The predictive distribution of each output is represented as

$$p(d_{j,*} \mid \tilde{\mathbf{z}}_*, \mathcal{D}_{\text{tr}}) = \mathcal{N}\left(\mu_{j,*}, \sigma_{\text{epi},j,*}^2 + \sigma_{n,j}^2\right). \tag{7.8}$$

Here, $\sigma_{\text{epi},j,*}$ is the posterior standard deviation of the latent function and decreases near informative data; $\sigma_{n,j}$ represents a learned constant dispersion. Thus, B2 provides total predictive uncertainty and an approximation of the epistemic component, but cannot represent local variations in aleatoric dispersion.

Both GPs are trained on the common partition, without additional data search or changes to normalization. Table 7.7 summarizes the specific decisions and times recorded in the training artifact.

Table 7.7: Model B2 configuration.

Item	Configuration
Role in the study	Homoscedastic probabilistic baseline
Model type	Two independent exact GPs
Kernel	Squared exponential with ARD
Outputs	Separate predictive distributions for $d_{v_y,k}$ and $d_{r,k}$
Noise model	Gaussian, learned, and constant for each output
Available uncertainty	Total and approximate latent epistemic
Training time	42.402 s for d_{v_y} and 35.840 s for d_r

The learned length scales are shown in Table 7.8. Because the inputs are standardized, a smaller scale indicates faster function variation with respect to that feature, but does not by itself constitute a causal measure of importance.

Table 7.8: ARD length scales learned by B2.

Input	GP for d_{v_y}	GP for d_r
V_x	2.6832	1.2561
v_y	3.1020	1.8271
r	0.9062	0.9574
δ	1.7534	1.3580

Table 7.9 presents predictive mean performance. Errors are close to those obtained by B1, allowing the probabilistic information from B2 to be studied without introducing

a relevant loss of nominal accuracy.

Table 7.9: B2 residual prediction performance at 20 ms.

	d_{v_y} (m/s)		d_r (rad/s)	
Partition	RMSE	MAE	RMSE	MAE
Training	0.009375	0.006769	0.007347	0.005371
Validation	0.008791	0.005860	0.006884	0.004620
ID test	0.010684	0.008057	0.008624	0.006552

On the ID test, the learned homoscedastic noise levels were 0.009575 m/s for d_{v_y} and 0.007583 rad/s for d_r. Table 7.10 summarizes the probabilistic diagnostics. Coverage of nominal 95 % Gaussian intervals is approximately 91.7 %, indicating slight overconfidence and motivating subsequent explicit calibration evaluation.

Table 7.10: Probabilistic diagnostics for B2 on the ID test.

Quantity	d_{v_y}	d_r
σ_n learned	0.009575 m/s	0.007583 rad/s
Mean predictive standard deviation	0.009824 m/s	0.007936 rad/s
Approximate mean epistemic standard deviation	0.001766 m/s	0.001841 rad/s
95 % interval coverage	91.68 %	91.78 %
Gaussian NLL	−3.1080	−3.3202

The one-step comparison in Table 7.11 confirms that the B2 mean retains the hybrid model's predictive improvement: it reduces the RMSE of v_y by 91.74 % and that of r by 90.16 % relative to B0. Its distinguishing contribution over B1 is therefore not a substantial improvement in the mean, but the availability of a homoscedastic predictive distribution that will serve as a reference for subsequent heteroscedastic models.

Table 7.11: One-step comparison between B0 and B2 on the ID test.

State	RMSE B0	RMSE B2	Improvement
v_y (m/s)	0.129336	0.010684	91.74 %
r (rad/s)	0.087671	0.008624	90.16 %

7.3.3 B3: Deep Ensemble of Heteroscedastic MLPs

B3 uses a *deep ensemble* of five heteroscedastic MLPs. Unlike B1, each member simultaneously predicts the mean and conditional variance of the two residual components. Unlike B2, aleatoric dispersion can vary with the input, and epistemic uncertainty is approximated through disagreement among independently trained networks.

Each member $m \in \{1, \ldots, M\}$ receives $\tilde{\mathbf{z}}_k$ and produces four outputs:

$$\left[\boldsymbol{\mu}_k^{(m)}, \mathbf{s}_k^{(m)}\right] = \mathbf{g}_{\boldsymbol{\theta}_m}(\tilde{\mathbf{z}}_k), \qquad \boldsymbol{\mu}_k^{(m)}, \mathbf{s}_k^{(m)} \in \mathbb{R}^2, \tag{7.9}$$

where $\boldsymbol{\mu}_k^{(m)}$ is the normalized residual mean and $\mathbf{s}_k^{(m)}$ represents log-variance. To maintain training stability, its components are restricted to $[-8, 5]$ and transformed through

$$\left(\boldsymbol{\sigma}^2\right)_k^{(m)} = \exp\left(\mathbf{s}_k^{(m)}\right). \tag{7.10}$$

Each network is fitted in normalized space by minimizing the heteroscedastic Gaussian negative log-likelihood (NLL). Omitting constants independent of the parameters, the loss for one member is

$$\mathcal{L}_{mathrmNLL}^{(m)} = \frac{1}{2N} \sum_{k=1}^{N} \sum_{j=1}^{2} \left[s_{j,k}^{(m)} + \exp\left(-s_{j,k}^{(m)}\right) \left(\tilde{d}_{j,k} - \mu_{j,k}^{(m)}\right)^2 \right]. \tag{7.11}$$

This function requires each member to balance mean error against the dispersion assigned to each sample; a large variance reduces the quadratic penalty but directly increases the log-variance term.

The joint prediction is obtained by combining the moments of the $M = 5$ members. For each residual component,

$$\bar{\mu}_k = \frac{1}{M} \sum_{m=1}^{M} \mu_k^{(m)}, \tag{7.12}$$

$$U_{\mathrm{epi},k}^2 = \frac{1}{M} \sum_{m=1}^{M} \left(\mu_k^{(m)} - \bar{\mu}_k\right)^2, \tag{7.13}$$

$$U_{\mathrm{ale},k}^2 = \frac{1}{M} \sum_{m=1}^{M} \left(\sigma^2\right)_k^{(m)}, \tag{7.14}$$

$$U_{\mathrm{tot},k} = \sqrt{U_{\mathrm{epi},k}^2 + U_{\mathrm{ale},k}^2}. \tag{7.15}$$

Means and standard deviations are then transformed into physical units using the common target statistics. U_{ale} must be interpreted as an operational estimate of learned conditional dispersion, not as a direct, pure measurement of physical noise.

Each MLP retains B1's two hidden layers of 64 neurons, but its output layer has four components. Training uses batches of 256 samples, a learning rate of 10^{-3}, a maximum of 250 epochs, and early stopping with a patience of 25 epochs. Base seed 42 controls reproducible member initialization; it does not define the data split, which remains fixed by Table 7.3. Training was performed on CPU. Table 7.12 summarizes these decisions.

All five members triggered early stopping and achieved similar validation NLL values, with no clearly anomalous network. Table 7.13 presents the final epoch and best validation

Table 7.12: Model B3 configuration.

Item	Configuration
Role in the study	Neural baseline with decomposed uncertainty
Model type	*Deep ensemble* of five heteroscedastic MLPs
Architecture per member	$4 \to 64 \to 64 \to 4$
Outputs per member	Two means and two log-variances
Loss function	Heteroscedastic Gaussian NLL
Batch size	256 samples
Learning rate	10^{-3}
Maximum epochs and patience	250 and 25, respectively
Log-variance bounds	$[-8, 5]$
Base seed	42
Available uncertainty	Epistemic, conditional aleatoric, and total

value for each member.

Table 7.13: Convergence of B3 ensemble members.

Member	Stopping epoch	Best validation NLL
1	187	−2.1428
2	122	−2.0985
3	169	−2.1416
4	120	−2.1212
5	145	−2.1013

On the ID test, the ensemble mean achieved the results in Table 7.14. Its accuracy remains on the same order as B1 and B2, so heteroscedastic estimation and ensemble disagreement are not obtained at the expense of relevant degradation in nominal prediction.

Table 7.14: B3 predictive performance on the ID test.

Metric	d_{v_y} (m/s)	d_r (rad/s)
RMSE	0.010591	0.008718
MAE	0.007949	0.006583

Table 7.15 shows that, within the nominal domain, mean aleatoric uncertainty is approximately five times greater than epistemic uncertainty. This is consistent with an ensemble whose members agree in regions represented by the data while maintaining residual conditional dispersion. Coverage remains below the nominal 95 %, so B3 is slightly overconfident on the ID test.

The preceding NLL corresponds to the moment-matched Gaussian approximation $\mathcal{N}(\bar{\mu}, U_{\text{epi}}^2 + U_{\text{ale}}^2)$; it is not the exact NLL of the five-Gaussian mixture. This convention allows comparison with B2, but must be retained when interpreting the values. Compared

Table 7.15: Probabilistic diagnostics for B3 on the ID test.

Quantity	d_{v_y}	d_r
U_{epi} (mean)	0.001680 m/s	0.001533 rad/s
U_{ale} (mean)	0.009230 m/s	0.007686 rad/s
U_{tot} (mean)	0.009438 m/s	0.007890 rad/s
95 % interval coverage	92.36 %	93.13 %
Moment-matched Gaussian NLL	−3.1869	−3.3687

with the homoscedastic GP, B3 modestly improves both coverage and NLL on the ID test, although neither model is perfectly calibrated.

Finally, Table 7.16 shows the comparison with B0. B3 reduces the RMSE of v_y by 91.81 % and that of r by 90.06 %. Its main contribution relative to B1 and B2 is to provide a competitive residual mean, conditional dispersion, and a model disagreement signal simultaneously.

Table 7.16: One-step comparison between B0 and B3 on the ID test.

State	RMSE B0	RMSE B3	Improvement
v_y (m/s)	0.129336	0.010591	91.81 %
r (rad/s)	0.087671	0.008718	90.06 %

7.3.4 B4: Two-Stage Heteroscedastic Gaussian Process

B4 retains B2's two mean GPs unchanged and adds a second stage to estimate input-dependent residual dispersion. This construction isolates the effect of heteroscedasticity: B2 and B4 share the same predictive mean and latent epistemic uncertainty, while differing in their treatment of residual dispersion, constant in B2 and variable in B4.

The first stage is defined by the exact GPs with squared-exponential ARD kernels described for B2. To train the second stage without using residuals obtained on the same samples used to fit the mean, out-of-fold predictions are generated within the training partition. If $g(i)$ identifies the group containing sample i, the corresponding residual is

$$e_{j,i}^{\text{OOF}} = d_{j,i} - \hat{\mu}_j^{(-g(i))}(\tilde{\mathbf{z}}_i), \qquad j \in \{v_y, r\}, \tag{7.16}$$

where $\hat{\mu}_j^{(-g(i))}$ is obtained without using the group of the evaluated sample. The RMSEs of these out-of-fold predictions were 0.010188 m/s for d_{v_y} and 0.008365 rad/s for d_r.

For each output, a second GP is fitted to a logarithmic transformation of the squared out-of-fold residual. The training artifact retains the correction $c_{\chi^2} = \mathbb{E}[\log(\chi_1^2)] = -1.27036$, used to compensate for the bias introduced by the logarithmic transforma-

tion. In compact form, this stage's target can be expressed as

$$q_{j,i} = \log\left[\left(e_{j,i}^{\mathrm{OOF}}\right)^2 + \epsilon_j\right] - c_{\chi^2}, \tag{7.17}$$

where $\epsilon_j > 0$ avoids the logarithm's singularity. If $\hat{q}_j(\tilde{\mathbf{z}}_*)$ is the dispersion GP prediction, the operational uncertainty components are obtained as

$$U_{\mathrm{ale},j,*}^2 = \exp[\hat{q}_j(\tilde{\mathbf{z}}_*)]\,, \tag{7.18}$$

$$U_{\mathrm{epi},j,*} = \sigma_{f,j,*}, \tag{7.19}$$

$$U_{\mathrm{tot},j,*} = \sqrt{U_{\mathrm{epi},j,*}^2 + U_{\mathrm{ale},j,*}^2}, \tag{7.20}$$

where $\sigma_{f,j,*}$ is the posterior standard deviation of the mean GP's latent function. These quantities are presented in physical units after reversing the common standardization.

The designation U_{ale} is retained to maintain a uniform taxonomy across models, but its interpretation requires caution. Because it is learned from e^{OOF}, this component may encompass irreducible noise, mean-model error, and variability between groups. In B4, it therefore represents a *heteroscedastic estimate of residual dispersion*, not a pure measurement of physical random noise.

Table 7.17 summarizes the decisions specific to B4. The data, frozen split, and standardization are those defined in the common environment and are therefore not repeated here.

Table 7.17: Model B4 configuration.

Item	Configuration
Role in the study	Heteroscedastic extension and direct ablation of B2
Mean stage	B2's two exact ARD-SE GPs, unchanged
Dispersion stage	Two independent GPs fitted to out-of-fold residuals
Second-stage target	Logarithm of the squared residual with correction $c_{\chi^2} = -1.27036$
Seed	42
U_{epi}	Latent posterior standard deviation of the mean GP
U_{ale}	Heteroscedastic estimate of residual dispersion
Total uncertainty	Quadratic combination of U_{epi} and U_{ale}

Because B4 retains B2's mean stage, both produce exactly the same point prediction errors on validation and the ID test. This equality provides an artifact consistency check and allows any probabilistic difference to be attributed exclusively to the second stage. Table 7.18 presents the corresponding metrics.

The diagnostics in Table 7.19 show that, on the ID test, residual dispersion dominates total uncertainty and has a magnitude similar to that estimated by B3, although the two models obtain it through different mechanisms. Coverage remains below the nominal

Table 7.18: B4 mean residual prediction performance at 20 ms.

	d_{v_y} (m/s)		d_r (rad/s)	
Partition	RMSE	MAE	RMSE	MAE
Validation	0.008791	0.005860	0.006884	0.004620
ID test	0.010684	0.008057	0.008624	0.006552

95 %, particularly for d_{v_y}, so B4 is also slightly overconfident on this partition.

Table 7.19: Probabilistic diagnostics for B4 on the ID test.

Quantity	d_{v_y}	d_r
U_{epi} (mean)	0.001766 m/s	0.001841 rad/s
U_{ale} (mean)	0.009608 m/s	0.007899 rad/s
U_{tot} (mean)	0.009873 m/s	0.008252 rad/s
95 % interval coverage	90.23 %	91.88 %
Gaussian NLL	−3.0422	−3.3273

On validation, coverage was 93.71 % for d_{v_y} and 95.16 % for d_r, with NLLs of −3.6606 and −3.7967, respectively. The degradation observed from validation to the ID test demonstrates that near-nominal calibration on validation does not guarantee that it will be maintained across groups not used during fitting.

The Spearman correlation between instantaneous uncertainties and absolute error is summarized in Table 7.20. U_{ale} exhibits a positive but weak association, whereas U_{epi} is practically uncorrelated with one-step error. Consequently, these signals should not be interpreted in isolation as direct predictors of the next error.

Table 7.20: Spearman correlation between uncertainty and absolute error for B4 on the ID test.

Indicator	$\lvert e_{v_y} \rvert$	$\lvert e_r \rvert$
U_{epi}	−0.0776	−0.0339
U_{ale}	0.1407	0.0849
U_{tot}	0.0425	0.0377

Finally, B4 reproduces the improvements of B2's mean over B0: it reduces the RMSE of v_y by 91.74 % and that of r by 90.16 %. Its contribution lies not in a new mean correction, but in replacing B2's constant dispersion with an input-dependent function. This relationship makes the B2–B4 comparison a direct ablation of the effect of modeling heteroscedasticity.

7.4 Distribution Shift Scenarios

The experimental campaign was organized into ten scenarios, E0–E9, designed to separate changes in excitation, observable variations in operating point, latent dynamics modifications, and changes occurring during execution. In every scenario, B0–B4 were evaluated on the same runs. B0 provides the purely physical reference; B1 measures the ability of a deterministic correction; and B2–B4 additionally allow the response of each model's available uncertainty components to be studied alongside predictive error. Learned parameters and normalization statistics remained frozen throughout the campaign; no E0–E9 scenario was used to retrain or readapt the models.

Vehicle A corresponds to the nominal configuration used to generate the training data. Vehicle B represents an alternative configuration with structural changes in properties such as mass, inertias, and tires. Longitudinal speed V_x is part of the residual models' input vector; in contrast, the grip coefficient μ and vehicle identity or parameters are not supplied as inputs. This difference allows observable covariate shifts to be contrasted with latent dynamic changes.

Table 7.21 summarizes the run configurations. E0 uses held-out DLC and DLCT runs not used during training and constitutes the in-distribution reference. The *single lane change* (SLC) maneuver is introduced as an excitation not observed during fitting. The remaining scenarios modify one or more conditions relative to reference E0.

Table 7.21: Configuration of the E0–E9 experimental campaign.

Reg.	Scenario	Maneuver or evaluation set	V_x (km/h)	μ	Vehicle
001	E0	DLC and DLCT; held-out runs	60	1.0	A
002	E1	SLC; new maneuver	60	1.0	A
003	E2	DLC and DLCT	50	1.0	A
004	E3	DLC and DLCT	70	1.0	A
005	E4	DLC and DLCT	60	0.8	A
006	E5	DLC and DLCT	60	0.6	A
007	E6	DLC and DLCT	60	0.3	A
008	E7	SLC; new maneuver	60	0.6	A
009	E8	DLC and DLCT	60	1.0	B
010	E9	Continuous DLC/DLCT	60	$1.0 \rightarrow 0.6$	A

Table 7.22 presents the type of change and experimental question associated with each scenario. The classification refers to the information available to the models, not only the magnitude of the physical modification.

7.4.1 Nominal Reference and Excitation Generalization: E0–E1

E0 establishes the performance level within the known distribution. Although it retains DLC and DLCT maneuvers, nominal speed, grip, and vehicle A, it uses complete runs

Table 7.22: Purpose of the experimental campaign scenarios.

Scenario	Applied change	Classification	Experimental question
E0	None	In-distribution reference	Do the learned models retain their predictive ability on nominal runs not used during training?
E1	SLC maneuver	Excitation generalization	Do they generalize to an unseen path when the physical configuration remains unchanged?
E2	$V_x \downarrow$	Observable covariate and operating point shift	Do error and U_{epi} increase when operating below the sole training speed?
E3	$V_x \uparrow$	Observable covariate and operating point shift	How do prediction and U_{epi} respond in a higher, more demanding regime?
E4	$\mu \downarrow$ moderate	Latent dynamic change	Do the uncertainties reflect degradation caused by an unobserved physical variable?
E5	$\mu \downarrow$ strong	Severe latent dynamic change	How do error and uncertainties evolve as the dynamic discrepancy increases?
E6	$\mu \downarrow$ extreme	Extreme dynamics test	Do confident failures occur, characterized by high error and low epistemic uncertainty?
E7	SLC maneuver and $\mu = 0.6$	Combined change	How do the models behave when unseen excitation is combined with a latent dynamic change?
E8	Vehicle B	Structural dynamics change	Do the uncertainties reflect changes in mass, inertias, and tires not included in the input vector?
E9	$\mu : 1.0 \rightarrow 0.6$ during the run	Dynamic change during execution	How do error and uncertainties evolve around the transition instant?

not involved in fitting. E0 therefore measures generalization across nominal runs, rather than performance on training samples. Its errors and uncertainties serve as the reference for interpreting all increases observed in E1–E9.

E1 retains $V_x = 60\,\mathrm{km/h}$, $\mu = 1$, and vehicle A, but replaces the identification maneuvers with a single SLC maneuver. This isolation evaluates generalization to an unseen excitation pattern without simultaneously introducing a change in physical parameters. The main comparison is E1 against E0.

7.4.2 Observable Operating Point Shifts: E2–E3

E2 and E3 evaluate extrapolation relative to the sole longitudinal speed present in training. E2 reduces V_x from 60 to 50 km/h, and E3 increases it to 70 km/h, retaining $\mu = 1$, vehicle A, and the DLC and DLCT maneuver families. Because V_x is an input to B1–B4, these scenarios constitute observable covariate shifts and allow checking whether the probabilistic models' epistemic uncertainty responds to departure from the nominal domain. Separate evaluation of E2 and E3 also identifies any asymmetry between extrapolating toward lower speeds and toward a more demanding regime.

7.4.3 Latent Dynamic Changes in Grip: E4–E6

E4–E6 form a sequence of increasing severity in which $V_x = 60\,\mathrm{km/h}$, vehicle A, and DLC and DLCT maneuvers are retained while μ decreases from its nominal value of 1.0 to 0.8, 0.6, and 0.3. Since μ is not part of the input vector, these scenarios modify the plant's input–output relationship without providing the model with a direct indication of the cause.

E4 represents a moderate perturbation; E5, a strong change; and E6, an extreme test. Joint analysis of the three scenarios determines whether error increases monotonically with the physical discrepancy and whether U_{epi}, U_{ale}, or U_{tot} track that increase. E6 is used specifically to search for confident failures: intervals in which error increases without a proportional rise in uncertainty.

7.4.4 Combined Maneuver and Grip Change: E7

E7 combines the two modifications previously isolated: it uses the unseen SLC maneuver and reduces grip to $\mu = 0.6$, retaining $V_x = 60\,\mathrm{km/h}$ and vehicle A. The natural references for interpretation are E1, which contains only the maneuver change, and E5, which contains only the same grip change. This comparison examines whether the joint degradation can be explained by the isolated changes or whether an additional interaction between excitation and dynamics appears.

7.4.5 Structural Vehicle Change: E8

E8 retains nominal maneuver, speed, and grip conditions, but replaces vehicle A with vehicle B. The scenario groups structural variations in mass, inertias, and tire properties that are not explicitly included in the learned models' input vector. Direct comparison with E0 studies sensitivity to a change of platform without confounding it with speed, surface, or maneuver variations. As in the grip scenarios, actual values of modified parameters are used to define and analyze the test, but are not provided as inputs to B1–B4.

7.4.6 Abrupt Change during Execution: E9

E9 introduces a temporal dimension absent from the preceding scenarios. During a continuous DLC/DLCT-based run, μ changes abruptly from 1.0 to 0.6, while $V_x = 60\,\mathrm{km/h}$ and vehicle A are retained. Rather than only summarizing performance over the entire run, this scenario is analyzed around the transition instant. The objective is to characterize the temporal evolution of error and uncertainties, including any delay between physical change, predictive degradation, and the response of available indicators. E0 and E5 provide the stationary references before and after the change, respectively.

7.5 Learned Model and Uncertainty

The structures, fitting criteria, and nominal-domain results of B1–B4 are presented in Section 7.3. At this methodological stage, the models are not retrained: they are used as frozen artifacts to generate the residual correction and available uncertainty signals on the same transitions of each scenario in Section 7.4. Thus, any difference between scenarios is attributed to the condition evaluated, rather than predictor readaptation.

For each model b, scenario s, and instant k, a common record is retained,

$$\mathcal{R}_k^{(b,s)} = \left\{ \mathbf{z}_k, \mathbf{d}_k, \bar{\mathbf{d}}_k^{(b)}, \mathbf{e}_{d,k}^{(b)}, \mathbf{U}_{\mathrm{epi},k}^{(b)}, \mathbf{U}_{\mathrm{ale},k}^{(b)}, \mathbf{U}_{\mathrm{tot},k}^{(b)}, S_{\mathrm{OOD},k}, t_{\mathrm{inf},k}^{(b)} \right\}. \tag{7.21}$$

The observed discrepancy $\mathbf{d}_k$ and error $\mathbf{e}_{d,k}^{(b)}$, defined according to Equation (4.13), are calculated after receiving the CarSim transition and used only as ground truth for evaluation. They are not among the signals available when taking the control action. Time $t_{\mathrm{inf},k}^{(b)}$ includes mean prediction and, where applicable, computation of its uncertainty components.

B1 provides only $\bar{\mathbf{d}}_k^{(1)}$ and acts as the deterministic reference; its uncertainty fields are recorded as unavailable, rather than zero. B2–B4 provide a mean and a probabilistic characterization, with the specific definitions in Sections 7.3.2, 7.3.3, and 7.3.4. This distinction avoids interpreting the absence of an estimate as perfect certainty. The general semantics of U_{epi}, U_{ale}, and U_{tot} are established in Section 4.2; differences between their

estimators remain visible throughout the analysis and are not presented as equivalent physical measurements.

The two residual outputs have different units and scales. Therefore, before forming scalar indicators, each uncertainty component is normalized by the target standard deviation calculated on the training partition, namely,

$$\widetilde{U}^{(b)}_{q,j,k} = \frac{U^{(b)}_{q,j,k}}{s_{d_j,\mathrm{tr}}}, \qquad \widetilde{U}^{(b)}_{q,k} = \sqrt{\frac{1}{2}\sum_{j\in\{v_y,r\}} \left(\widetilde{U}^{(b)}_{q,j,k}\right)^2}, \quad q \in \{\mathrm{epi}, \mathrm{ale}, \mathrm{tot}\}. \tag{7.22}$$

Componentwise signals are retained for diagnostics; aggregation is used only when the supervisor requires a scalar input. This prevents an output from dominating the combination because of its unit or numerical magnitude. Since B2–B4 construct uncertainty through different mechanisms, their main comparison is performed both in absolute terms and relative to each model's E0 reference.

Calibration is evaluated without using E1–E9 to modify the predictive distributions. If a recalibration transformation is applied, its parameters are estimated only on the validation partition described in Section 7.2, after fixing the predictor configuration, and remain frozen during the ID test and shift campaign. Coverage and NLL are calculated per output and scenario according to Equations (4.6) and (4.7). This separation checks whether calibration obtained under nominal conditions is retained, becomes conservative, or produces overconfidence under each family of change.

An OOD score computed in standardized input space is used as complementary information. The reference option is a regularized Mahalanobis distance relative to the training data:

$$S_{\mathrm{OOD},k} = \sqrt{\left(\widetilde{\mathbf{z}}_k - \boldsymbol{\mu}_{\widetilde{z},\mathrm{tr}}\right)^{\mathsf{T}} \left(\boldsymbol{\Sigma}_{\widetilde{z},\mathrm{tr}} + \lambda \mathbf{I}\right)^{-1} \left(\widetilde{\mathbf{z}}_k - \boldsymbol{\mu}_{\widetilde{z},\mathrm{tr}}\right)}, \qquad \lambda > 0. \tag{7.23}$$

The mean, covariance, regularization, and any threshold are determined without using the test scenarios and then kept fixed. The inverse matrix can be computed before operation, so online evaluation requires only operations on four inputs. This score detects departure in $\mathbf{z}_k$, but is not assumed capable of revealing latent dynamic changes on its own, such as a grip variation that does not shift the observed covariates.

In E0–E8, analysis is performed per run and then aggregated by scenario, avoiding treating correlated transitions as independent repetitions. In E9, time series synchronized with the change instant are also retained. Any smoothing used by the supervisor must be causal, depend only on the current and previous samples, and retain the same parameters across all scenarios. Post-change error is reserved as an evaluation label and is never used to anticipate the uncertainty or OOD signal.

Finally, the computational cost of each block is recorded through the distribution of

$t_{\text{inf},k}^{(b)}$ and compared with sampling period T_s. This check is performed with the models already loaded and, when necessary, separates the time spent on the mean, uncertainty, and OOD score. Only signals computable before applying $\mathbf{u}_k$, calibrated without test-scenario information, and compatible with the execution deadline proceed to the construction of C_k.

7.6 Construction and Evaluation of the Validity Signal

7.7 Use of the Validity Signal in MPC/NMPC

7.8 Reference Architectures and Ablation Study

B0 – Physical Model Only:
Controller using a physical model without a learned residual.

B1 – Learned Model without Validity Monitoring:
Controller using a physical model and a learned residual, without validity estimation or out-of-distribution detection.

B2 – Validity/Uncertainty Only:
Controller using a learned model and a signal constructed from uncertainty and calibration, without an explicit out-of-distribution detection component.

B3 – Out-of-Distribution Detection Only:
Controller using a learned model and a response based solely on the OOD score, without probabilistic uncertainty estimation.

P – Proposed Approach:
Controller using a physical model, a learned residual, and the signal

$$C_k = g(U_{\text{epi}}, U_{\text{ale}}, \text{Cal}, OOD)$$

to adapt the MPC/NMPC controller.

7.9 Evaluation Metrics

Metrics are first calculated for each complete run and then aggregated by scenario. Consecutive transitions are not considered independent repetitions because they share a trajectory, initial state, and temporal evolution. When two architectures are evaluated on the same CarSim run, the comparison is paired. Results are presented per component

and family of change; a single average over E0–E9 is used only as a secondary summary, since it could hide failures concentrated in a particular regime.

7.9.1 Model Metrics

One-step evaluation uses the residual error in Equation (4.13). For component $j \in \{v_y, r\}$, model b, and run q with n_q transitions, the following are calculated:

$$\mathrm{RMSE}_j^{(b,q)} = \sqrt{\frac{1}{n_q}\sum_{k=1}^{n_q}\left(e_{d,j,k}^{(b)}\right)^2}, \tag{7.24}$$

$$\mathrm{MAE}_j^{(b,q)} = \frac{1}{n_q}\sum_{k=1}^{n_q}\left|e_{d,j,k}^{(b)}\right|. \tag{7.25}$$

RMSE penalizes large errors more strongly, while MAE checks that the comparison does not depend on a few extreme samples. Both are retained in physical units. To quantify the residual's contribution relative to B0, the relative RMSE reduction is also reported,

$$I_{\mathrm{RMSE},j}^{(b,q)} = 100\left(1 - \frac{\mathrm{RMSE}_j^{(b,q)}}{\mathrm{RMSE}_j^{(\mathrm{B0},q)}}\right), \tag{7.26}$$

expressed as a percentage. A positive value represents improvement, and a negative value indicates that the learned correction harms prediction. This measure complements the instantaneous gain defined in Equation (4.14) and allows comparison of the residual's average usefulness across scenarios.

Multistep behavior is evaluated using the state error in Equation (4.15). For a horizon H, the following is used:

$$E_H^{(b,q)} = \sqrt{\frac{1}{H}\sum_{i=1}^{H}\left\|\mathbf{e}_{x,i|k}^{(b,q)}\right\|_{\mathbf{W}_x}^2}, \tag{7.27}$$

where $\mathbf{W}_x$ contains fixed scales to prevent states with different units or magnitudes from dominating the result. Errors are also reported per state and step i, because the aggregate value does not show whether degradation appears at the beginning or end of the horizon. To isolate model quality, B0–B4 receive the same input sequence in this test; differences caused by the actions selected by each controller are subsequently evaluated in closed loop.

7.9.2 Uncertainty, Calibration, and Out-of-Distribution Detection Metrics

The probabilistic quality of B2–B4 is characterized through the coverage and NLL defined in Equations (4.6) and (4.7). B3's moment-matching convention is retained across all

scenarios so that its values are comparable with B2 and B4. Because coverage at a single level may favor unnecessarily wide intervals, several nominal levels $\mathcal{P} = \{p_1, \ldots, p_L\}$ are evaluated, and the mean calibration error is calculated as

$$\mathrm{CE}_j^{(b,s)} = \frac{1}{L} \sum_{\ell=1}^{L} \left| \widehat{\mathrm{Cov}}_j^{(b,s)}(p_\ell) - p_\ell \right|. \tag{7.28}$$

This result is accompanied by mean interval width. This distinguishes a well-calibrated, sharp distribution from one that obtains coverage through excessive dispersion.

The usefulness of uncertainties as error indicators is evaluated separately for U_{epi}, U_{ale}, and U_{tot}. Spearman correlation with $|e_{d,j,k}|$ is reported both per output and for the dimensionless quantities in Equation (7.22). Correlation is complemented by the confident failure rate,

$$\mathrm{FCR}_j^{(b,s)} = \frac{\sum_k \mathbb{I}\left\{ |e_{d,j,k}^{(b)}| > \tau_{e,j},\ U_{\mathrm{tot},j,k}^{(b)} \le \tau_{U,j} \right\}}{\sum_k \mathbb{I}\left\{ |e_{d,j,k}^{(b)}| > \tau_{e,j} \right\}}, \tag{7.29}$$

which represents the fraction of high errors accompanied by uncertainty that is not high. Thresholds $\tau_{e,j}$ and $\tau_{U,j}$ are fixed using quantiles of the nominal validation partition and are not retuned for each scenario. If no error exceeds $\tau_{e,j}$, the rate is declared undefined rather than assigned a zero.

The score in Equation (7.23) is evaluated using E0 as the negative reference and each shift scenario as the positive set. AUROC, AUPR, and the false-positive rate at the selected sensitivity level are reported. These metrics are broken down by scenario: combining them into a single OOD class would hide the difference between observable covariate shifts and latent dynamics changes. For the latter, a low AUROC does not necessarily contradict the functioning of the input score; it indicates that the change is not identifiable through $\mathbf{z}_k$ and must be considered when assessing complementarity with uncertainties.

A temporal evaluation is added for E9. Given a causal indicator a_k, a threshold γ_a, and a minimum persistence of L_a samples, detection delay is

$$T_{\mathrm{det}}^{(a)} = T_s \left(\min \left\{ k \ge k_{\mathrm{shift}} : a_{k:k+L_a-1} > \gamma_a \right\} - k_{\mathrm{shift}} \right). \tag{7.30}$$

The threshold and persistence are fixed before evaluating E9. Alongside delay, the false-alarm rate in the pre-change segment and the proportion of runs with no detection are reported. Thus, a rapid response is not rewarded if it comes at the expense of frequent alarms.

The execution cost of monitoring is summarized through the median, 95th percentile, and maximum of the times recorded in Equation (7.21). The deadline miss rate is also

calculated,

$$R_{\text{miss}} = \frac{1}{N} \sum_{k=1}^{N} \mathbb{I}\{t_{\text{total},k} > T_s\}\,, \tag{7.31}$$

where $t_{\text{total},k}$ includes prediction, uncertainty, OOD, and validity signal construction. Time is measured after loading the models, without mixing initialization with periodic inference.

7.9.3 Closed-Loop Performance Metrics

Closed-loop metrics constitute the main criterion for determining whether validity estimation provides value to the controller. The RMSE and maximum absolute values of e_y and e_ψ are calculated for each run. The former measures sustained behavior, while the latter preserves transient errors that might disappear in the average. Dynamic states v_y and r are summarized through RMS and absolute maxima to check that improved tracking does not come at the expense of excessive lateral responses.

Overall performance is also expressed through a realized cost using the same weights as the predictive problem in Equation (3.19):

$$J_{\text{cl}}^{(q)} = \frac{1}{n_q} \sum_{k=0}^{n_q-1} \left(\left\| \mathbf{x}_k - \mathbf{x}_k^{\text{ref}} \right\|_{\mathbf{Q}}^2 + \left\| \mathbf{u}_k \right\|_{\mathbf{R}}^2 \right). \tag{7.32}$$

This cost facilitates scalar comparison, but does not replace reporting its components. Steering effort is characterized through the RMS and maximum of δ_k, and its variation through

$$\text{TV}_{\delta}^{(q)} = \sum_{k=1}^{n_q-1} |\delta_k - \delta_{k-1}|. \tag{7.33}$$

This latter metric detects oscillations or control activity that an average cost might conceal.

Constraints are analyzed without reducing them to a binary variable. For each scalar constraint written as $g_c(\mathbf{x}_k, \mathbf{u}_k) \leq 0$, the fraction of violating samples, maximum magnitude, and accumulated violation are recorded,

$$V_c^{(q)} = \sum_{k=0}^{n_q-1} \max\{0, g_c(\mathbf{x}_k, \mathbf{u}_k)\}\, T_s. \tag{7.34}$$

The proportion of runs with at least one violation is also reported. This prevents many successful runs from hiding a severe failure under a specific condition.

Computational feasibility is evaluated through the rate of successfully solved optimization problems, the number of fallback action activations, median solution time, its 95th percentile and maximum, and the rate of cycles exceeding T_s. These indicators are presented alongside horizon N_k, so that any tracking improvement can be related to the

cost of MPC/NMPC adaptation.

Post-change recovery is also measured in E9. Recovery time is the interval from k_{shift} until tracking error again remains within a band fixed from E0 for L_{rec} samples. If the run ends before satisfying the criterion, it is recorded as not recovered; it is not replaced with the maximum observed duration. The maximum post-change error and accumulated cost during recovery are also reported.

The primary outcomes are the RMSE and maximum of e_y, constraint violations, optimizer feasibility, and compliance with the sampling deadline. Heading error, dynamic states, control effort, steering variation, and recovery time serve as complementary outcomes. This hierarchy is fixed before the final comparison to avoid selecting only metrics favorable to an architecture after the fact.

7.10 Empirical Robustness Criterion

$$\Delta J_{\text{prop}} < \Delta J_{\text{ref}}, \tag{7.35}$$

7.11 Level of Guarantee and Scope of the Conclusions

7.12 Statistical Analysis and Reproducibility

Chapter 8

Preliminary Results

8.1 Feasibility of Learning Vehicle Dynamics Models

8.2 Response of Uncertainty and Out-of-Distribution Indicators to Distribution Shifts

8.3 Preliminary Integration with Control

8.4 Discussion of Preliminary Results

Chapter 9

Expected Contributions

9.1 Control-Relevant Validity Estimation

9.2 Validity-Aware Vehicle Control

9.3 Runtime Validity Characterization

9.4 Experimental Taxonomy of Shifts in Vehicle Dynamics

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Chapter 10

Work Plan (Priority 1)

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10.2 Schedule

10.3 Decision Milestones

Chapter 11

Risks, Limitations, and Mitigation Strategies

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11.2 Risk of Inadequate Calibration or Weak Correlation with Error

11.3 Risk That Horizon Adaptation Does Not Improve Closed-Loop Performance

11.4 Computational Risk

11.5 Risk of Scope Expansion

11.6 Limitations of Formal Guarantees

Chapter 12

Conclusions of the Qualifying Examination

Bibliography

[1] Alberto Bertipaglia, Mohsen Alirezaei, Riender Happee, and Barys Shyrokau. Model predictive contouring control for vehicle obstacle avoidance at the limit of handling. In Wei Huang and Mehdi Ahmadian, editors, *Advances in Dynamics of Vehicles on Roads and Tracks III*, volume 2 of *Lecture Notes in Mechanical Engineering*, pages 132–142. Springer, 2024.

[2] Alberto Bertipaglia, Mohsen Alirezaei, Riender Happee, and Barys Shyrokau. An unscented Kalman filter-informed neural network for vehicle sideslip angle estimation. *IEEE Transactions on Vehicular Technology*, 73(9):12731–12746, 2024.

[3] Bosch. 30 years of ESP: technology developed by Bosch evolves and continues to shape the future of mobility. Bosch Media Service Brasil, 2025. Press release in Portuguese; title translated into English. Accessed October 12, 2026.

[4] Bosch. Safe braking. Bosch Global, n.d. Accessed October 13, 2026.

[5] Lukas Brunke, Melissa Greeff, Adam W. Hall, Zhaocong Yuan, Siqi Zhou, Jacopo Panerati, and Angela P. Schoellig. Safe learning in robotics: From learning-based control to safe reinforcement learning. *Annual Review of Control, Robotics, and Autonomous Systems*, 5:411–444, 2022.

[6] Guoying Chen, Chenxiao Bi, Xuanming Zhao, Xinyu Wang, Zhenhai Gao, Wei Deng, Huili Yu, and Liunan Yang. Data-mechanism fused integrated motion control for distributed vehicle considering online model mismatch compensation. *IEEE Transactions on Vehicular Technology*, 75(9):19147–19160, 2026.

[7] Yutao Chen, Jie He, Jun Cheng, Jingli Huang, and Jie Huang. Hybrid offline–online Gaussian process regression-based model predictive control for autonomous vehicles trajectory tracking. *Engineering Applications of Artificial Intelligence*, 182:115841, 2026.

[8] Shuo Cheng, Bin-Bin Hu, Heng-Lai Wei, Liang Li, and Chen Lv. Deep learning-based hybrid dynamic modeling and improved handling stability assessment for autonomous vehicles at driving limits. *IEEE Transactions on Vehicular Technology*, 74(4):5582–5593, 2025. Published online December 11, 2024.

[9] J. Hu, H. Liu, S. Yi, and C. Huang. Lane-changing control strategy for distributed drive vehicles considering yaw stability. *IET Intelligent Transport Systems*, 19:e12605, 2025.

[10] Miren Isasa, Ángela Alonso-Solórzano, Itziar Gurrutxaga, and Heriberto Pérez-Acebo. A skid resistance predicting model for single carriageways. *Lubricants*, 13(8):365, 2025.

[11] Juraj Kabzan, Lukas Hewing, Alexander Liniger, and Melanie N. Zeilinger. Learning-based model predictive control for autonomous racing. *IEEE Robotics and Automation Letters*, 4(4):3363–3370, 2019.

[12] Dvij Kalaria, Qin Lin, and John M. Dolan. Adaptive planning and control with time-varying tire models for autonomous racing using extreme learning machine. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 10443–10449, 2024.

[13] Lun Li, Deping Wang, Zhicheng Chen, Bing Zhu, Jian Zhao, Haiqiao Li, Jiayi Han, Dongjian Song, and Peixing Zhang. Trajectory-tracking control of autonomous vehicles with dynamic nonlinear coupling characteristics and parameter uncertainty. *International Journal of Automotive Technology*, 27:1035–1049, 2026. Published online September 1, 2025.

[14] Wei Pan, Haoqi Hu, Yiding Hua, Han Wang, Lin Zhang, and Hong Chen. A real-time nonlinear robust MPC strategy for electric vehicle stability with model uncertainty. *Control Engineering Practice*, 172:106894, 2026.

[15] S. Salleh, R. Muhamad, and M.H. Abdillah. Performance of ralumac micro surfacing at blackspot eastbound, latar highway. *IOP Conference Series: Materials Science and Engineering*, 527(1):012054, 2019.

[16] Viktor Skrickij, Paulius Kojis, Eldar Šabanovič, Barys Shyrokau, and Valentin Ivanov. Review of integrated chassis control techniques for automated ground vehicles. *Sensors*, 24(2):600, 2024.

[17] Dongxu Su, Zhiguo Zhao, and Kun Zhao. Reinforcement learning-based tube nonlinear model predictive control for robust trajectory tracking control of autonomous vehicles. *IEEE Transactions on Vehicular Technology*, 75(9):18857–18870, 2026.

[18] Wenjie Wang, Hao Chen, Ran Shu, Solyeon Kwon, Kyoungseok Han, and Hongyu Shu. Adaptive deep Koopman operator for vehicle dynamics modeling: A physics-informed and tire-force-driven approach. *Expert Systems with Applications*, 333:134228, 2027. Available online August 29, 2026.

[19] Zihang Weng, Zhen Leng, Yuchuan Du, Chenglong Liu, Xinqiang Zhang, Danning Li, and Yulin He. Leveraging tabular prior-data fitted networks for accurate pavement friction coefficient prediction from 3d texture features. *Tribology International*, 218:111794, 2026.

[20] Feixiang Xu, Junkang Feng, Yafei Wang, Xiaoyi Wang, Chuanwang Shen, and Chen Zhou. Parallel self-learning adaptive strategy based on model predictive control for trajectory tracking of autonomous vehicles. *Engineering Applications of Artificial Intelligence*, 173:114495, 2026.

[21] Kumlachew Yeneneh and Gadisa Sufe. AutoDyn: robust model predictive control design for trajectory tracking under varying vehicle dynamics. *Scientific Reports*, 16:25130, 2026.

[22] Kumlachew Yeneneh, Bisrat Yoseph, and Gadisa Sufe. Adaptive model predictive control for robust lateral motion tracking of semi-autonomous vehicles with dynamic parameter variation. *Scientific Reports*, 16:716, 2026. Published online November 30, 2025.

[23] Duk Sun Yun and Byung Chul Lim. Real-time axle-load sensing and AI-enhanced braking-distance prediction for multi-axle heavy-duty trucks. *Applied Sciences*, 16:1547, 2026.

[24] Zhisong Zhou, Yafei Wang, Xulei Liu, Zexing Li, Mingyu Wu, and Guofeng Zhou. Hybrid of neural network and physics-based estimator for vehicle longitudinal dynamics modeling using limited driving data. *IEEE Transactions on Intelligent Transportation Systems*, 26(10):16735–16746, 2025.